

Chapter 11

Cell Communication

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

Neil Campbell and Jane Reece

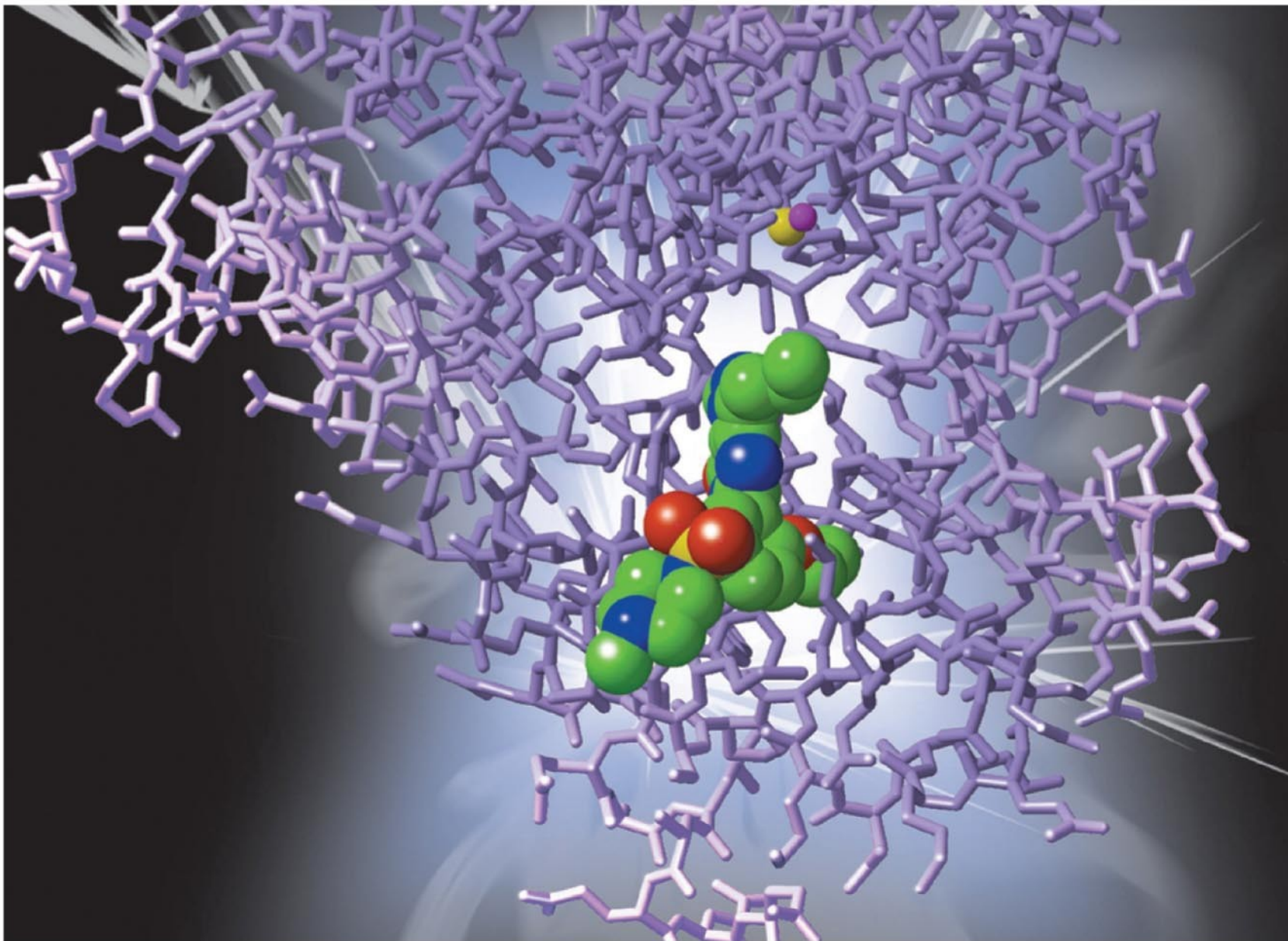
Lectures by Chris Romero, updated by Erin Barley with contributions from Joan Sharp

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Overview: The Cellular Internet

- Cell-to-cell communication is essential for multicellular organisms
- Biologists have discovered some universal mechanisms of cellular regulation
- The combined effects of multiple signals determine cell response
- For example, the dilation of blood vessels is controlled by multiple molecules

Fig. 11-1



Concept 11.1: External signals are converted to responses within the cell

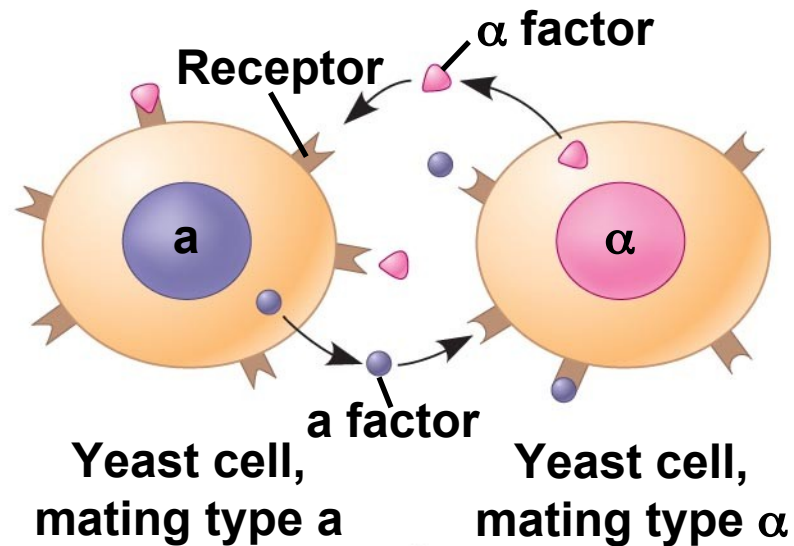
- Microbes are a window on the role of cell signaling in the evolution of life

Evolution of Cell Signaling

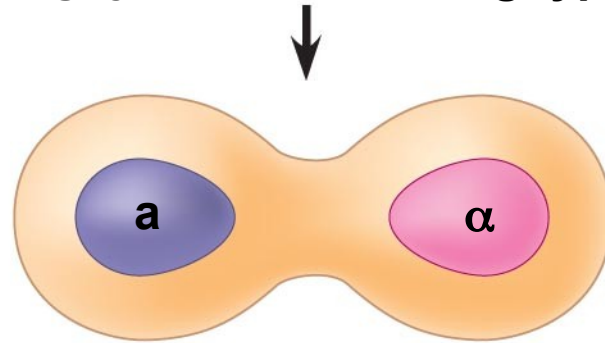
- A **signal transduction pathway** is a series of steps by which a signal on a cell's surface is converted into a specific cellular response
- Signal transduction pathways convert signals on a cell's surface into cellular responses

Fig. 11-2

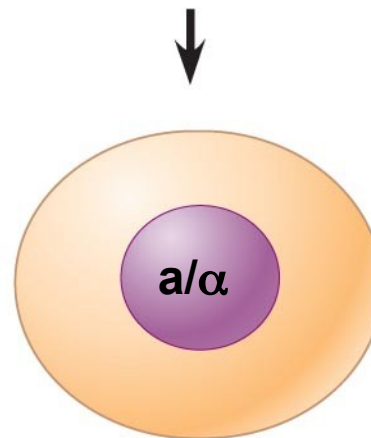
1 Exchange of mating factors



2 Mating



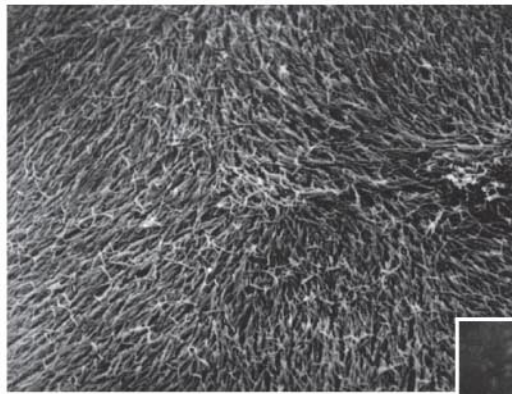
3 New a/α cell



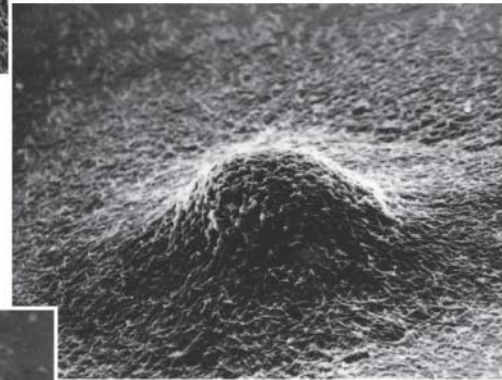
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- Pathway similarities suggest that ancestral signaling molecules evolved in prokaryotes and were modified later in eukaryotes
 - The concentration of signaling molecules allows bacteria to detect population density

Fig. 11-3

1 Individual rod-shaped cells

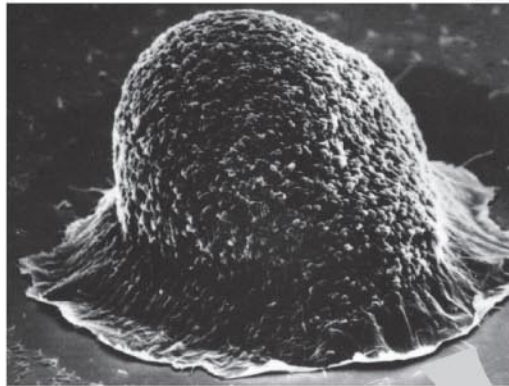


2 Aggregation in process

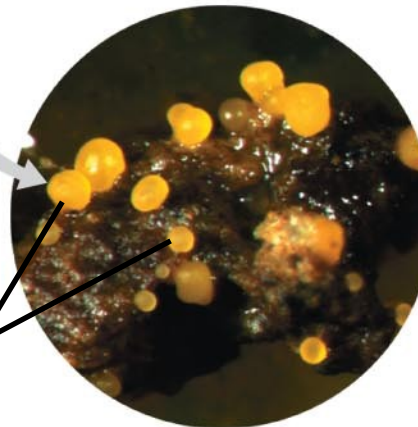


0.5 mm

3 Spore-forming structure (fruiting body)

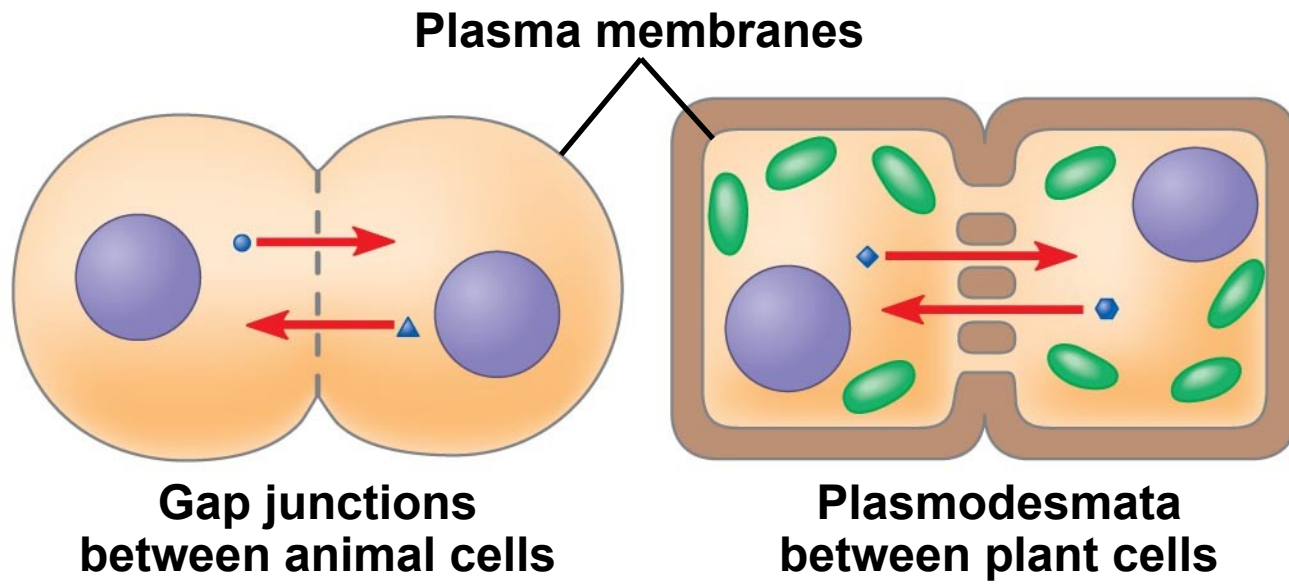


Fruiting bodies

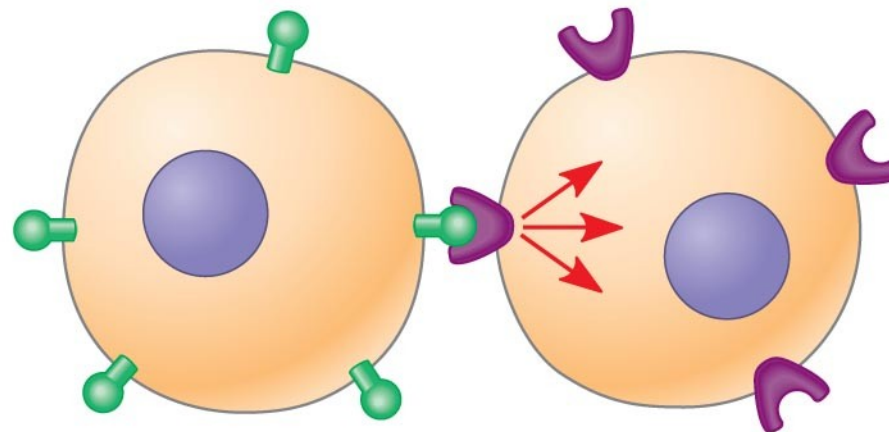


Local and Long-Distance Signaling

- Cells in a multicellular organism communicate by chemical messengers
- Animal and plant cells have cell junctions that directly connect the cytoplasm of adjacent cells
- In local signaling, animal cells may communicate by direct contact, or cell-cell recognition



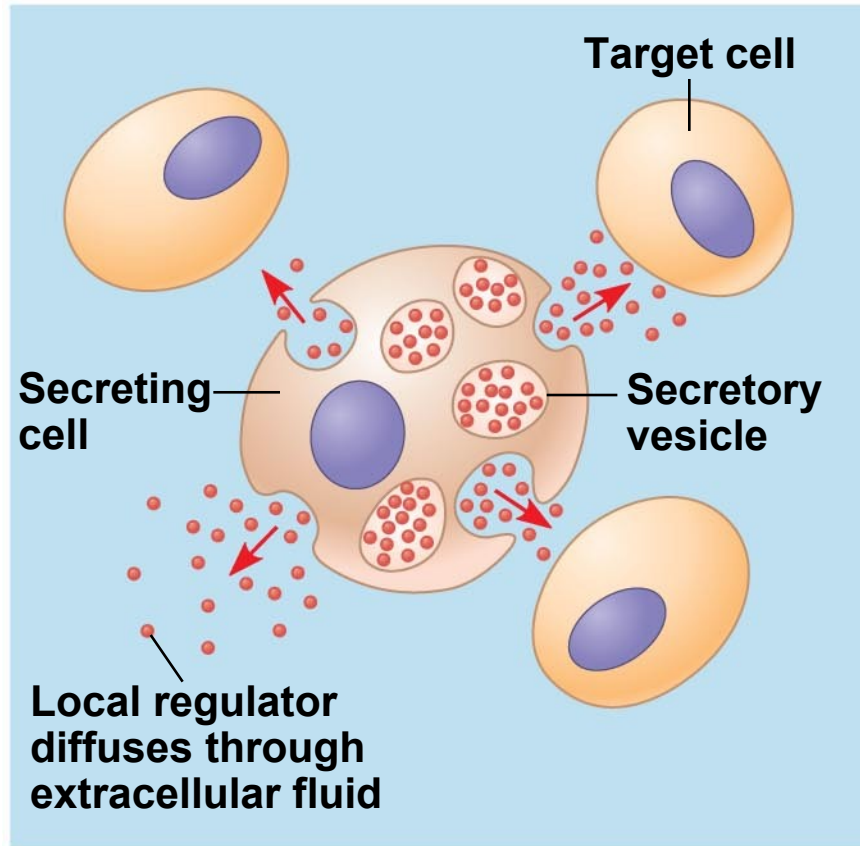
(a) Cell junctions



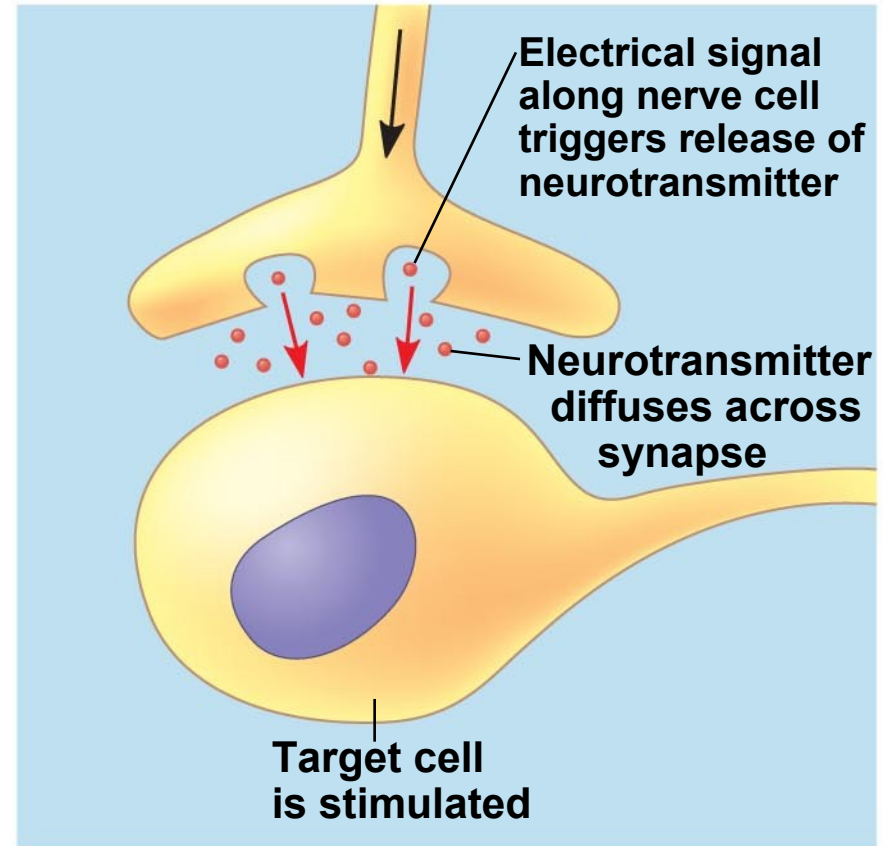
(b) Cell-cell recognition

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- In many other cases, animal cells communicate using **local regulators**, messenger molecules that travel only short distances
 - In long-distance signaling, plants and animals use chemicals called **hormones**

Local signaling

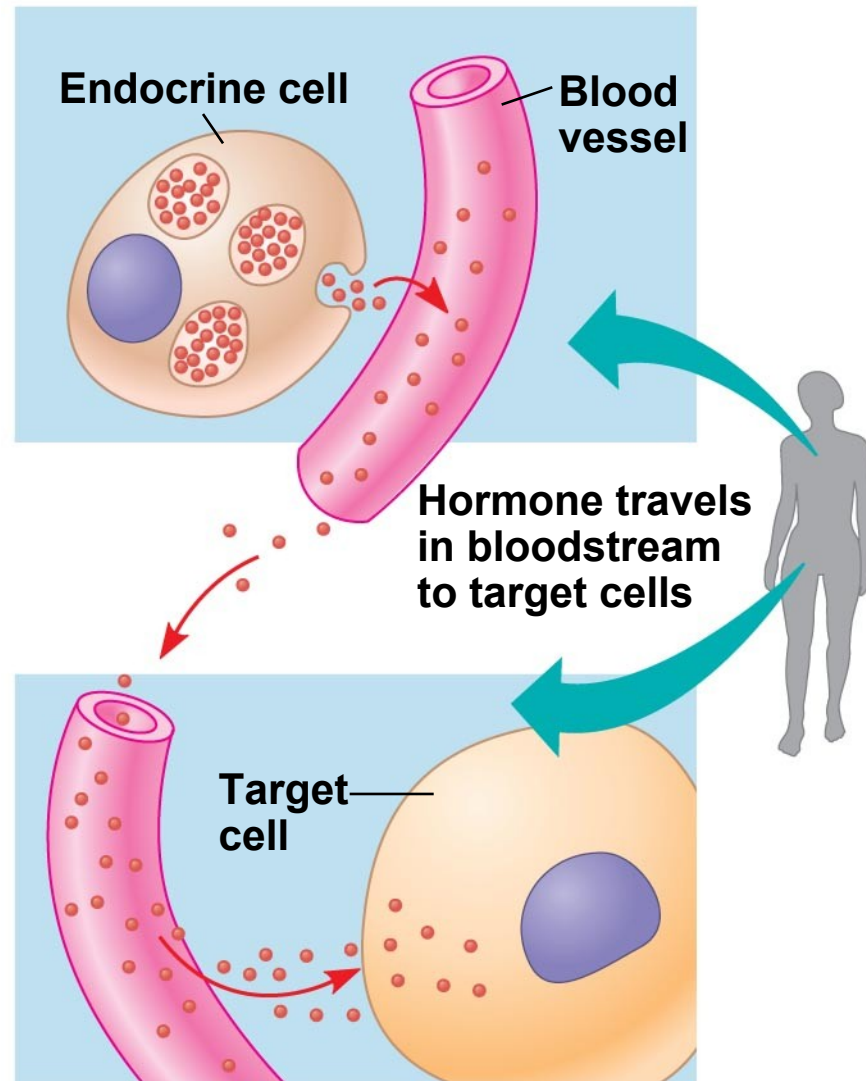


(a) Paracrine signaling



(b) Synaptic signaling

Long-distance signaling



(c) Hormonal signaling

The Three Stages of Cell Signaling: *A Preview*

- Earl W. Sutherland discovered how the hormone epinephrine acts on cells
- Sutherland suggested that cells receiving signals went through three processes:
 - **Reception**
 - **Transduction**
 - **Response**

Fig. 11-6-1

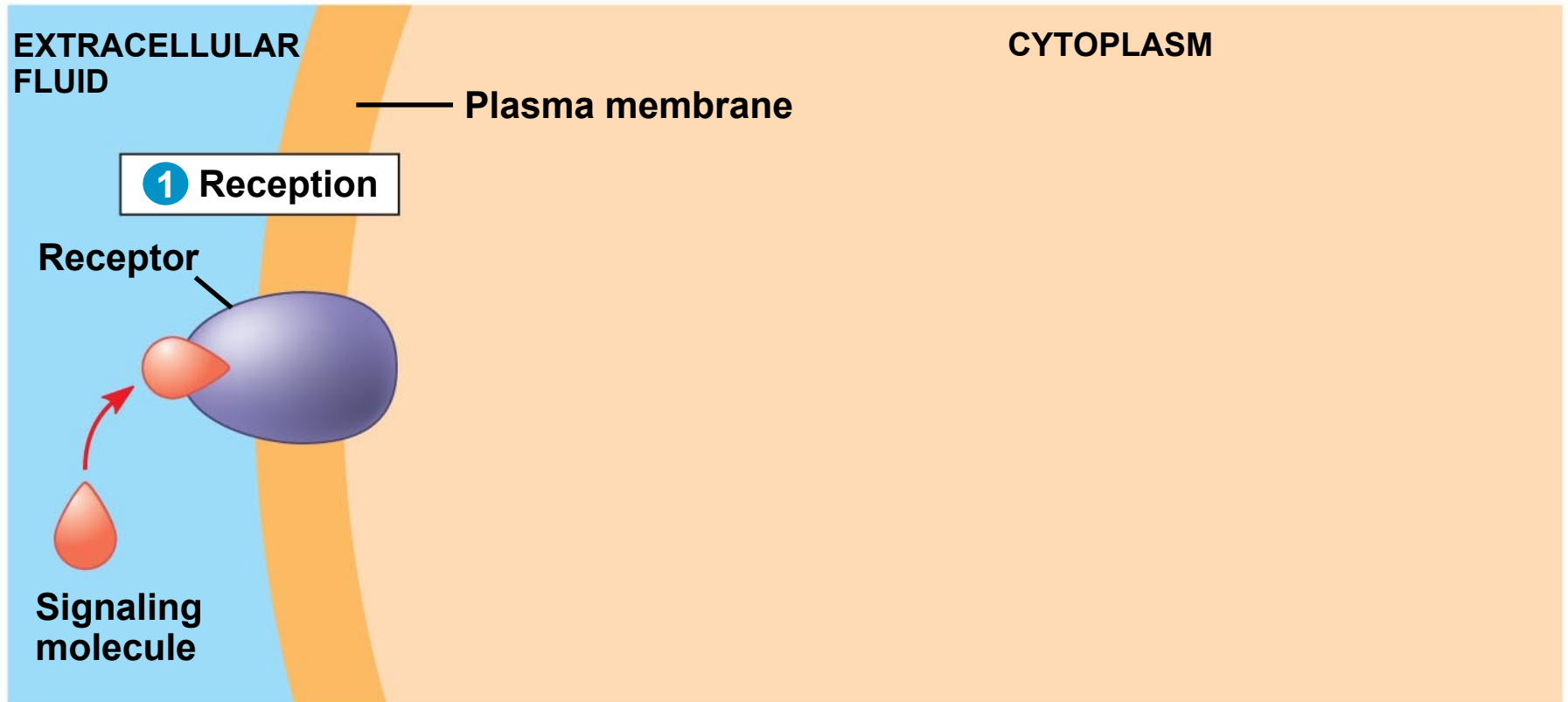


Fig. 11-6-2

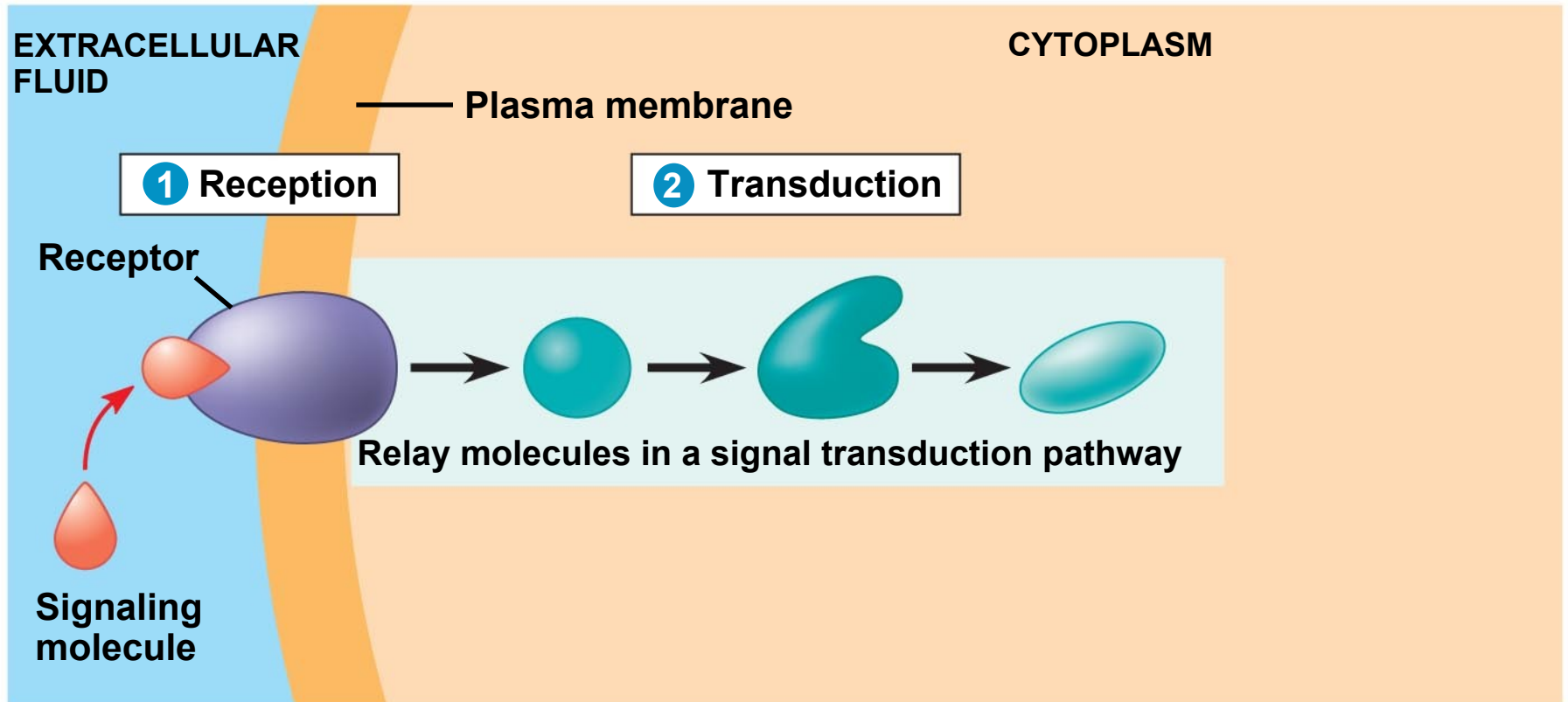
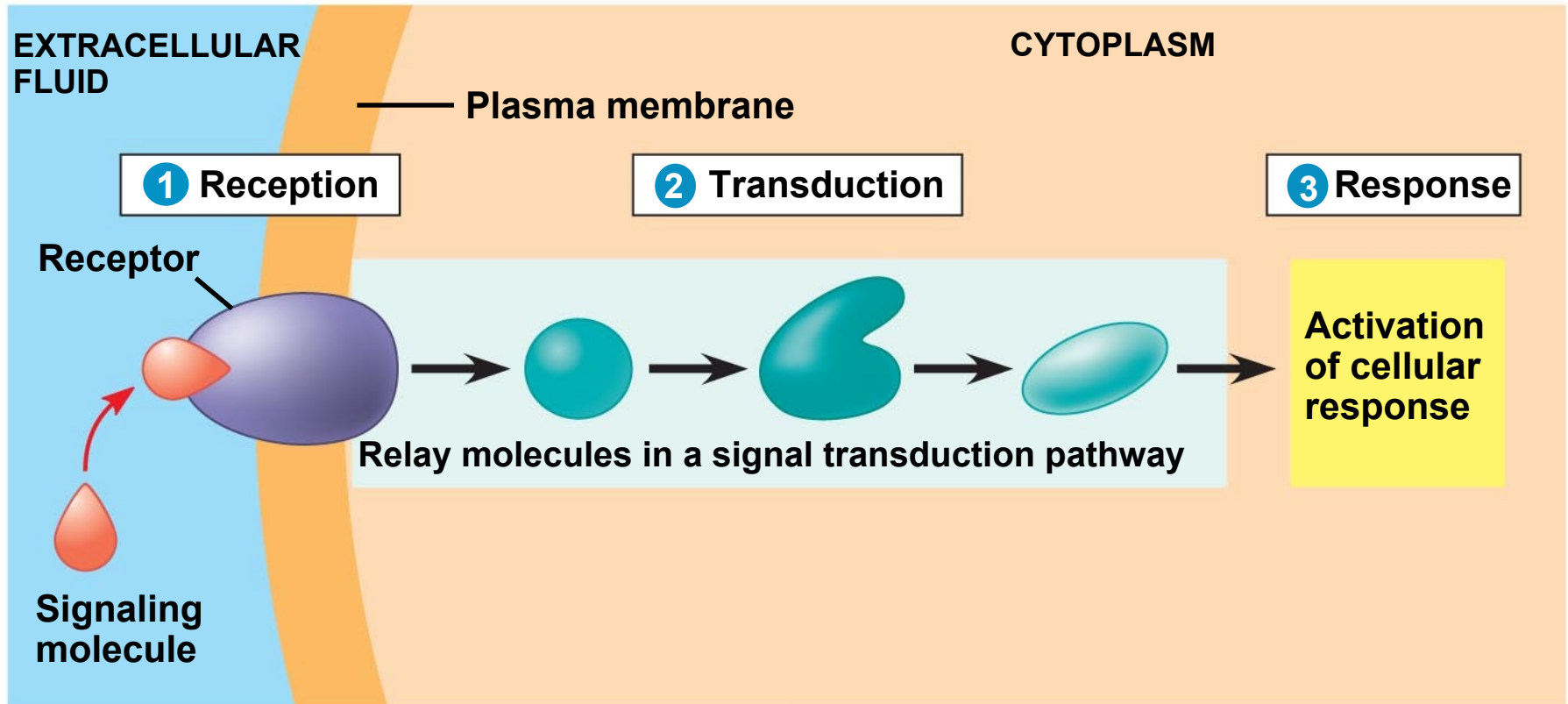


Fig. 11-6-3



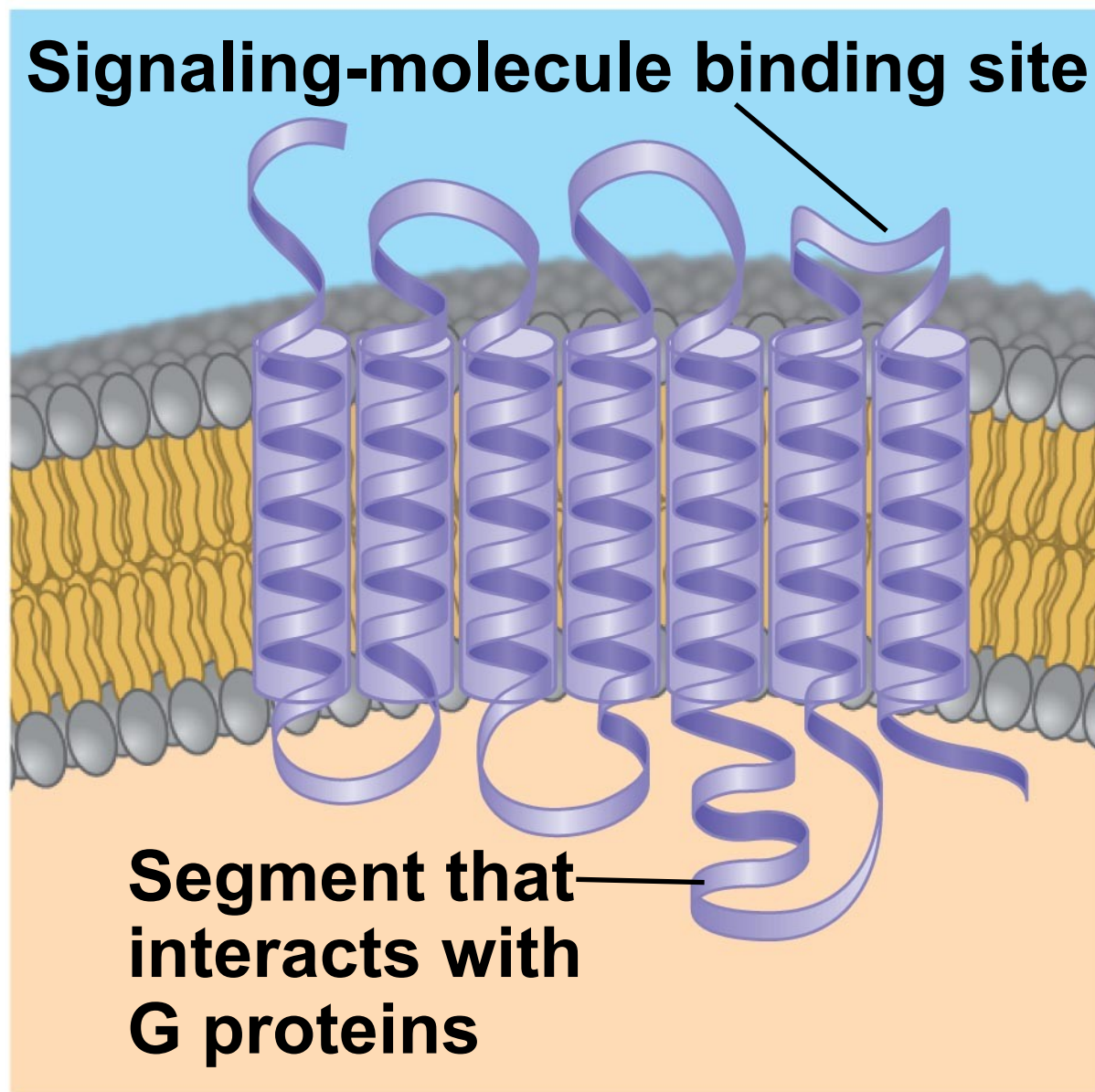
Concept 11.2: Reception: A signal molecule binds to a receptor protein, causing it to change shape

- The binding between a signal molecule (**ligand**) and receptor is highly specific
- A shape change in a receptor is often the initial transduction of the signal
- Most signal receptors are plasma membrane proteins

Receptors in the Plasma Membrane

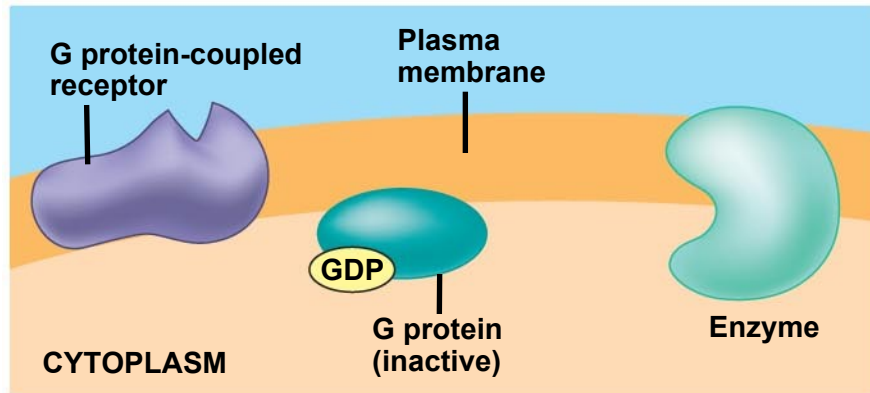
- Most water-soluble signal molecules bind to specific sites on receptor proteins in the plasma membrane
- There are three main types of membrane receptors:
 - G protein-coupled receptors
 - Receptor tyrosine kinases
 - Ion channel receptors

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- A **G protein-coupled receptor** is a plasma membrane receptor that works with the help of a **G protein**
 - The G protein acts as an on/off switch: If GDP is bound to the G protein, the G protein is inactive

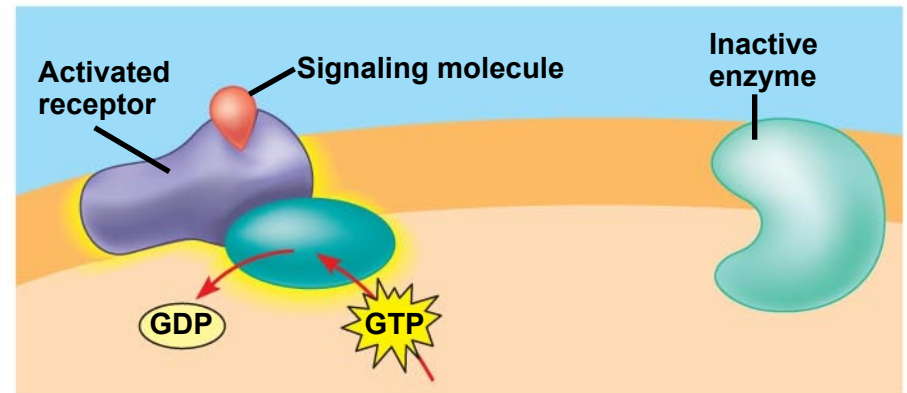


G protein-coupled receptor

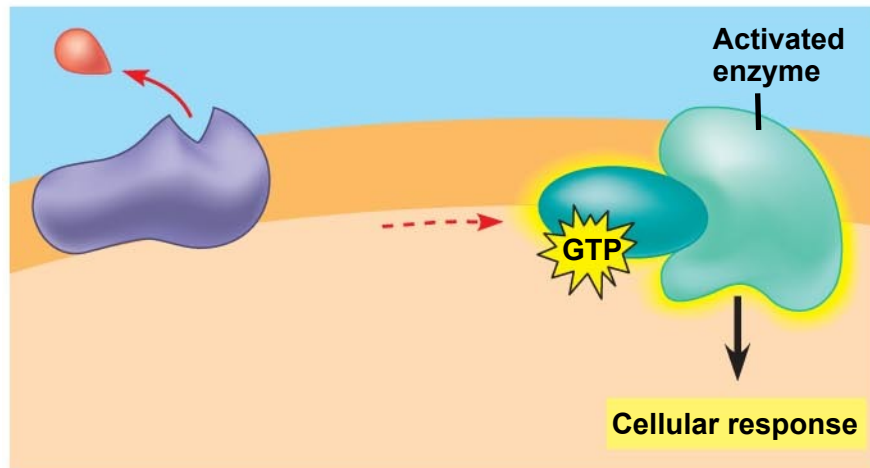
Fig. 11-7b



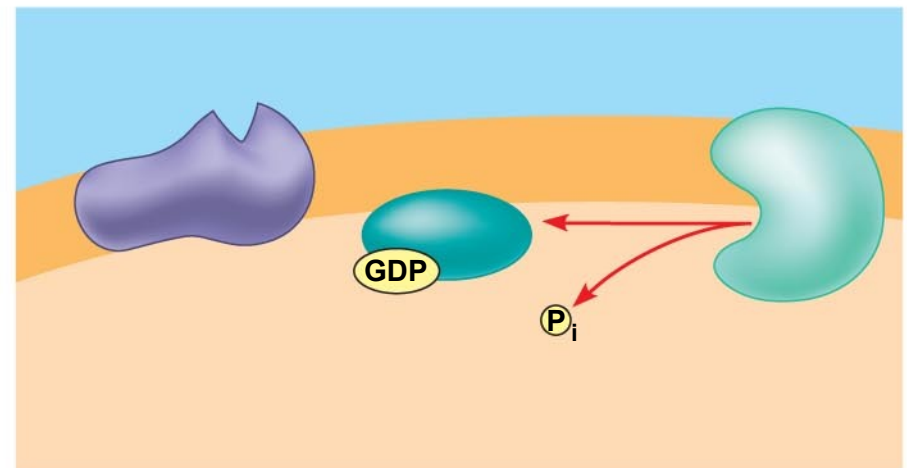
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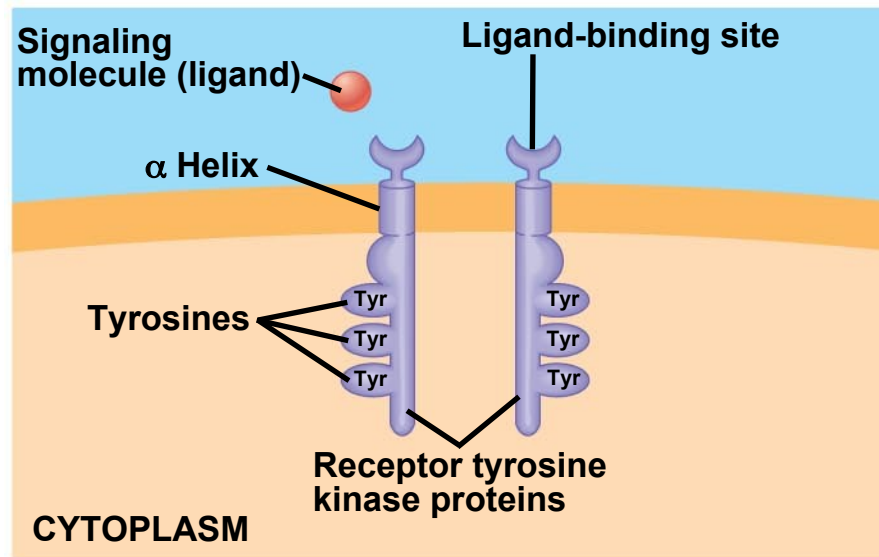
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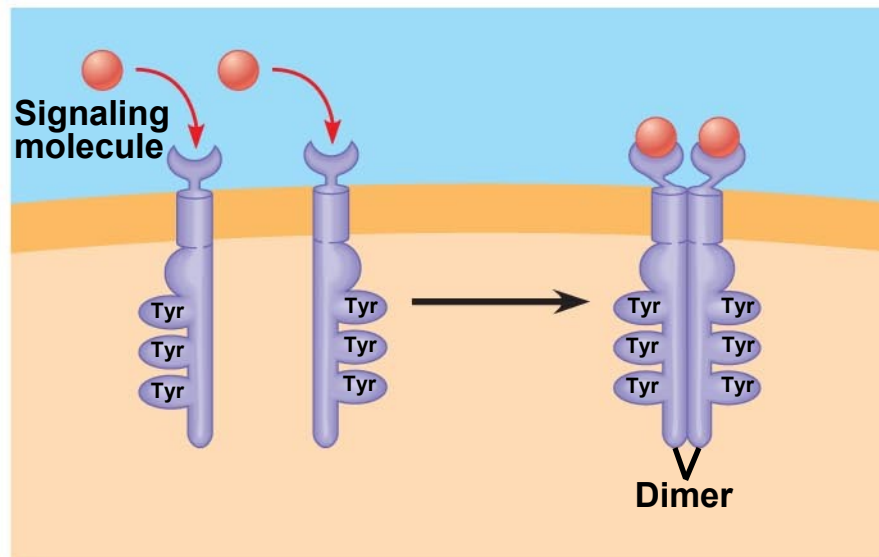
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- **Receptor tyrosine kinases** are membrane receptors that attach phosphates to tyrosines
 - A receptor tyrosine kinase can trigger multiple signal transduction pathways at once

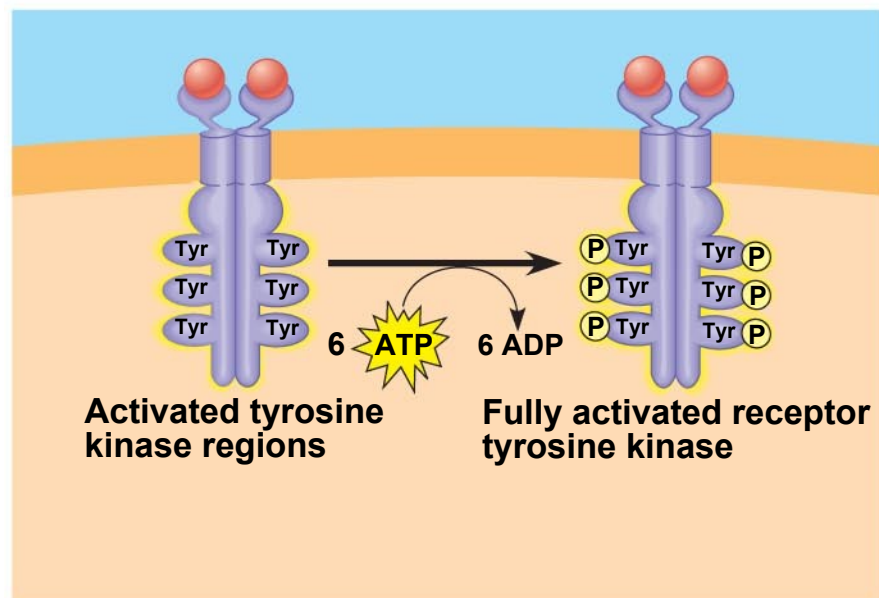
Fig. 11-7c



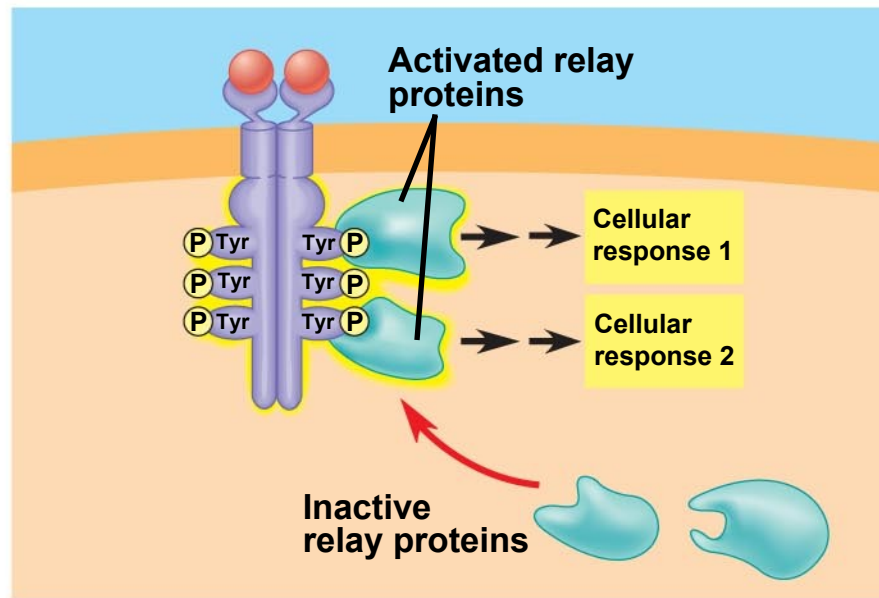
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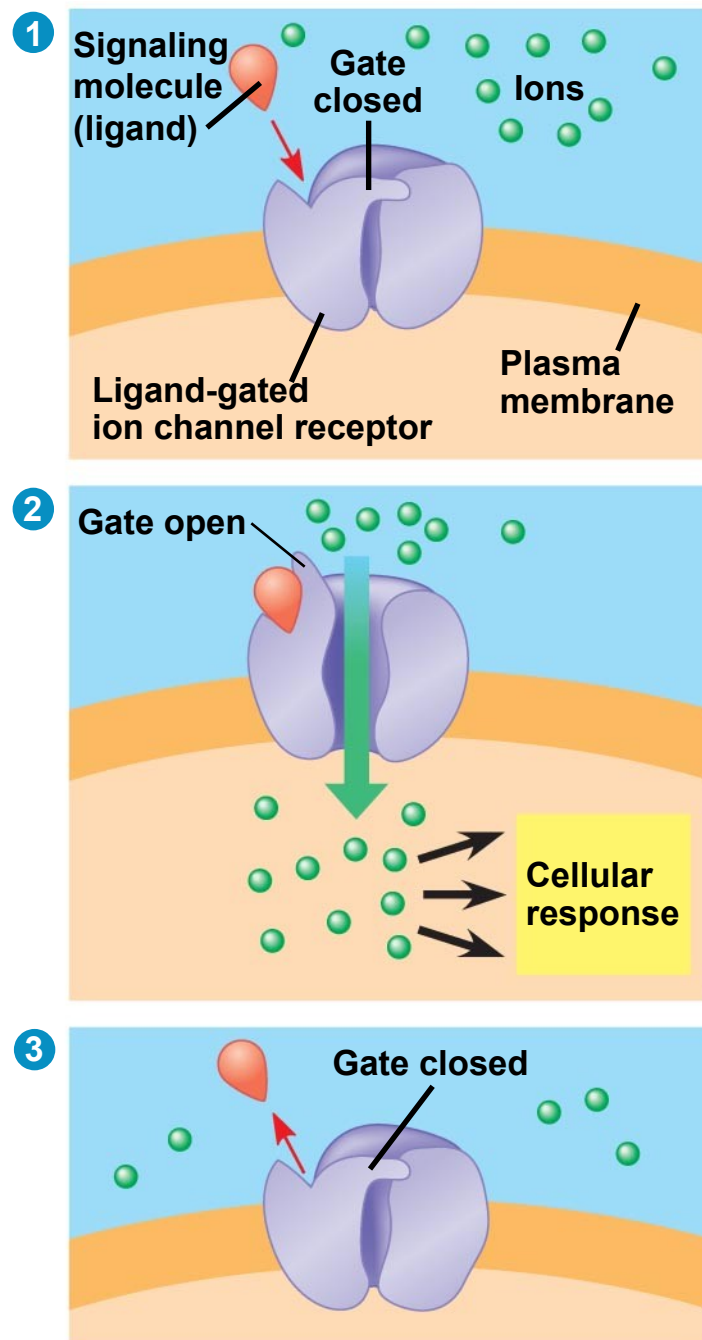
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4

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- A **ligand-gated ion channel** receptor acts as a gate when the receptor changes shape
 - When a signal molecule binds as a ligand to the receptor, the gate allows specific ions, such as Na^+ or Ca^{2+} , through a channel in the receptor

Fig. 11-7d



Intracellular Receptors

- Some receptor proteins are intracellular, found in the cytosol or nucleus of target cells
- Small or hydrophobic chemical messengers can readily cross the membrane and activate receptors
- Examples of hydrophobic messengers are the steroid and thyroid hormones of animals
- An activated hormone-receptor complex can act as a transcription factor, turning on specific genes

Fig. 11-8-1

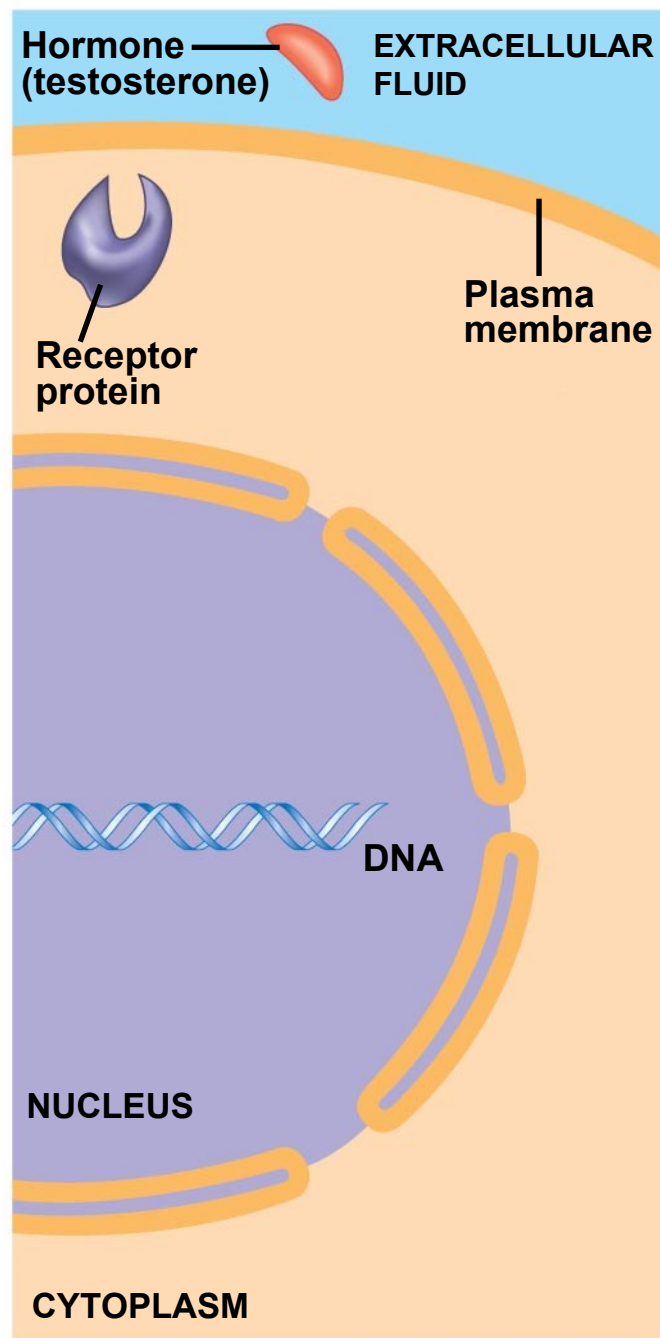


Fig. 11-8-2

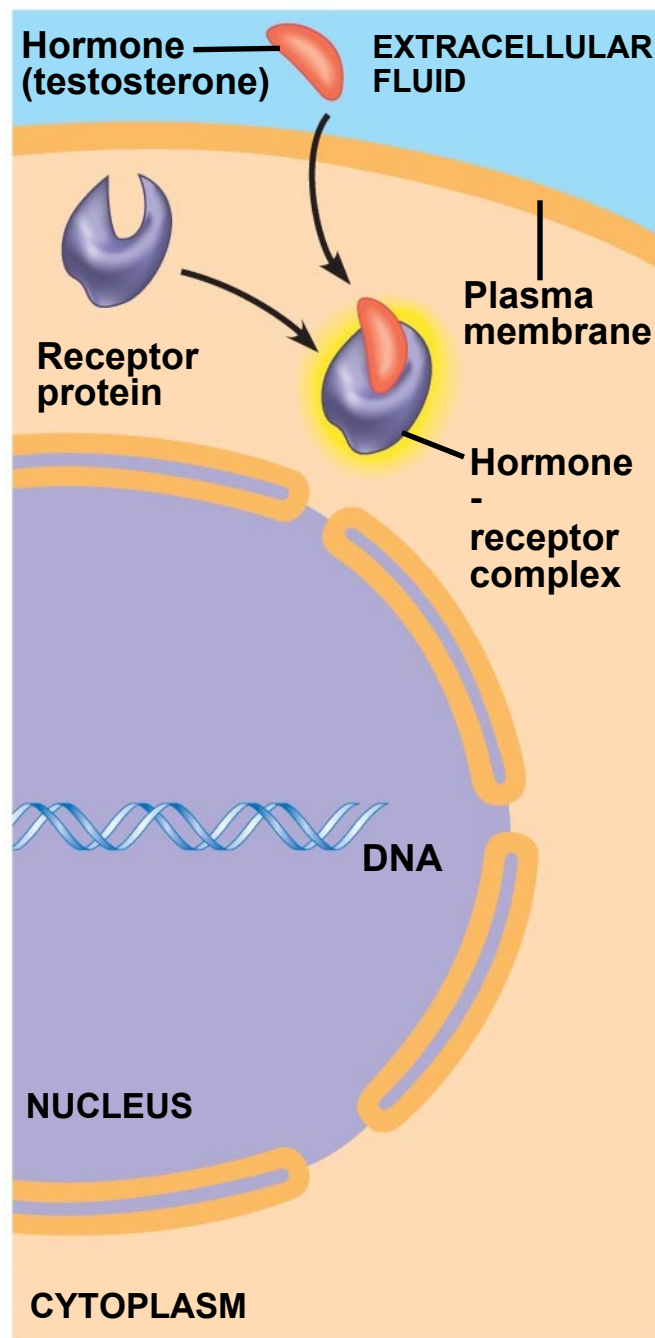


Fig. 11-8-3

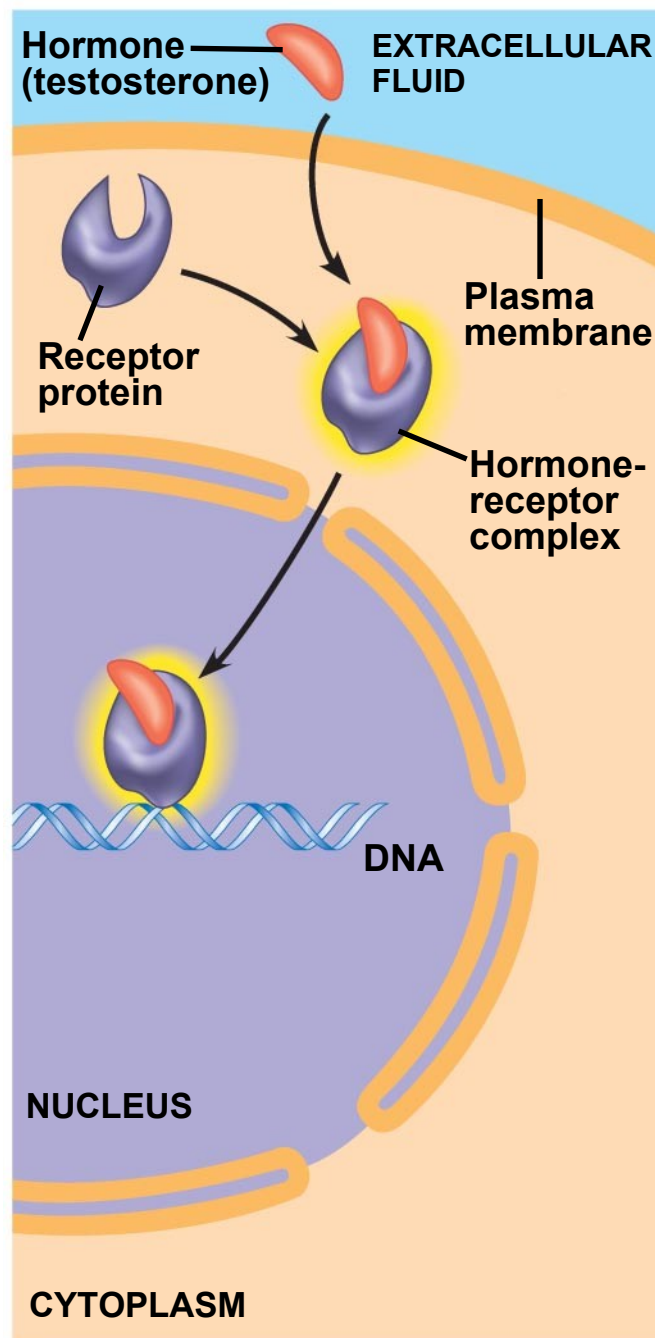


Fig. 11-8-4

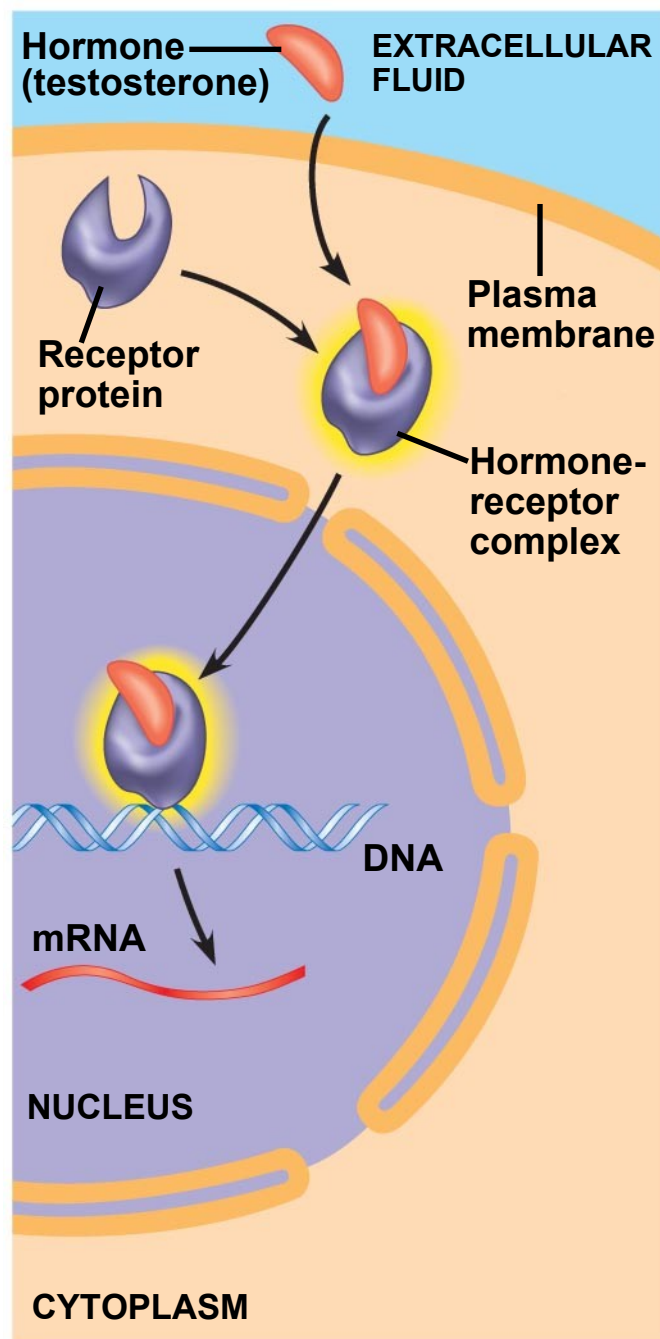
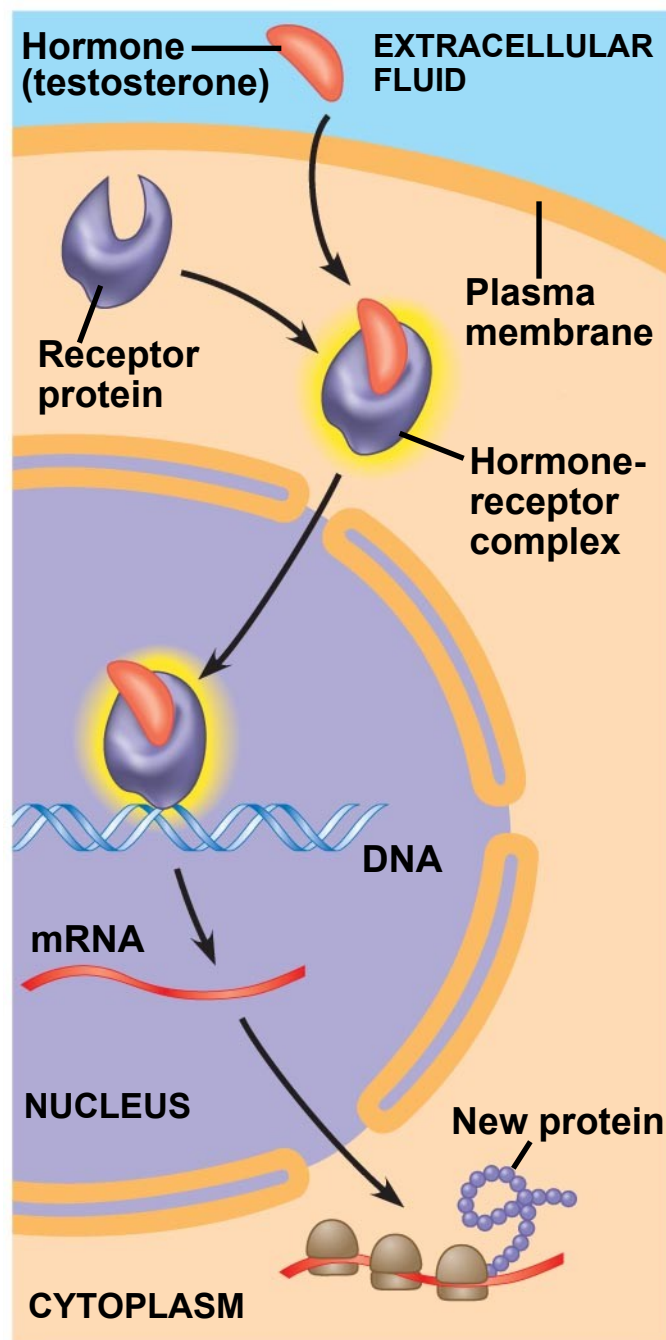


Fig. 11-8-5



Concept 11.3: Transduction: Cascades of molecular interactions relay signals from receptors to target molecules in the cell

- Signal transduction usually involves multiple steps
- Multistep pathways can amplify a signal: A few molecules can produce a large cellular response
- Multistep pathways provide more opportunities for coordination and regulation of the cellular response

Signal Transduction Pathways

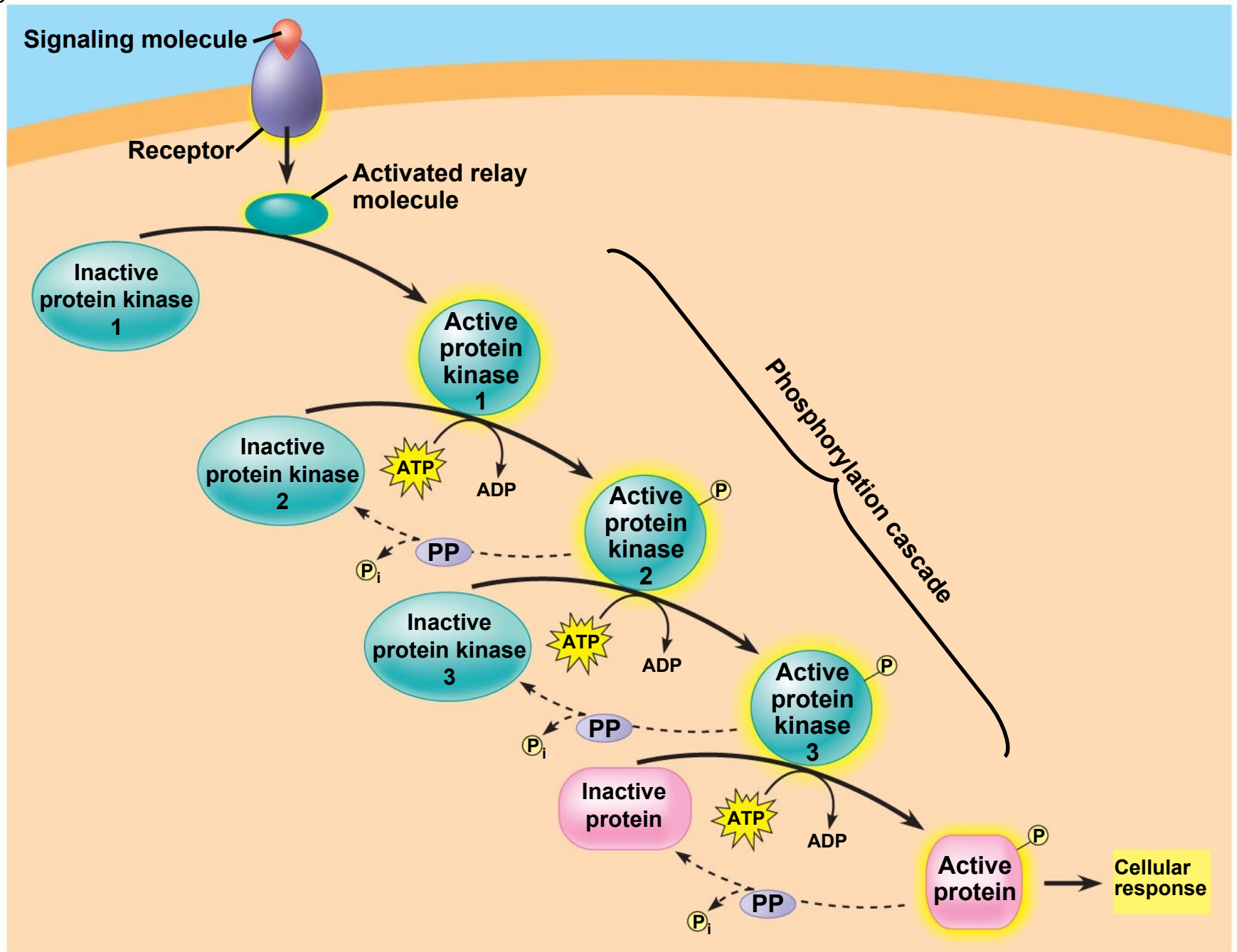
- The molecules that relay a signal from receptor to response are mostly proteins
- Like falling dominoes, the receptor activates another protein, which activates another, and so on, until the protein producing the response is activated
- At each step, the signal is transduced into a different form, usually a shape change in a protein

Protein Phosphorylation and Dephosphorylation

- In many pathways, the signal is transmitted by a cascade of protein phosphorylations
- **Protein kinases** transfer phosphates from ATP to protein, a process called phosphorylation

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- **Protein phosphatases** remove the phosphates from proteins, a process called dephosphorylation
 - This phosphorylation and dephosphorylation system acts as a molecular switch, turning activities on and off

Fig. 11-9



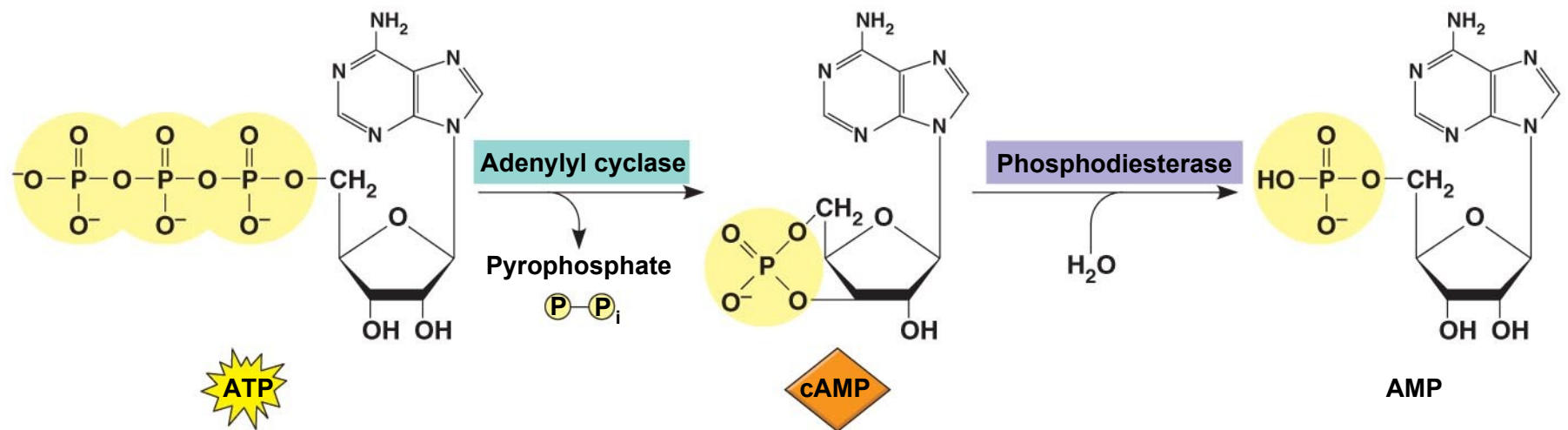
Small Molecules and Ions as Second Messengers

- The extracellular signal molecule that binds to the receptor is a pathway's “first messenger”
- **Second messengers** are small, nonprotein, water-soluble molecules or ions that spread throughout a cell by diffusion
- Second messengers participate in pathways initiated by G protein-coupled receptors and receptor tyrosine kinases
- Cyclic AMP and calcium ions are common second messengers

Cyclic AMP

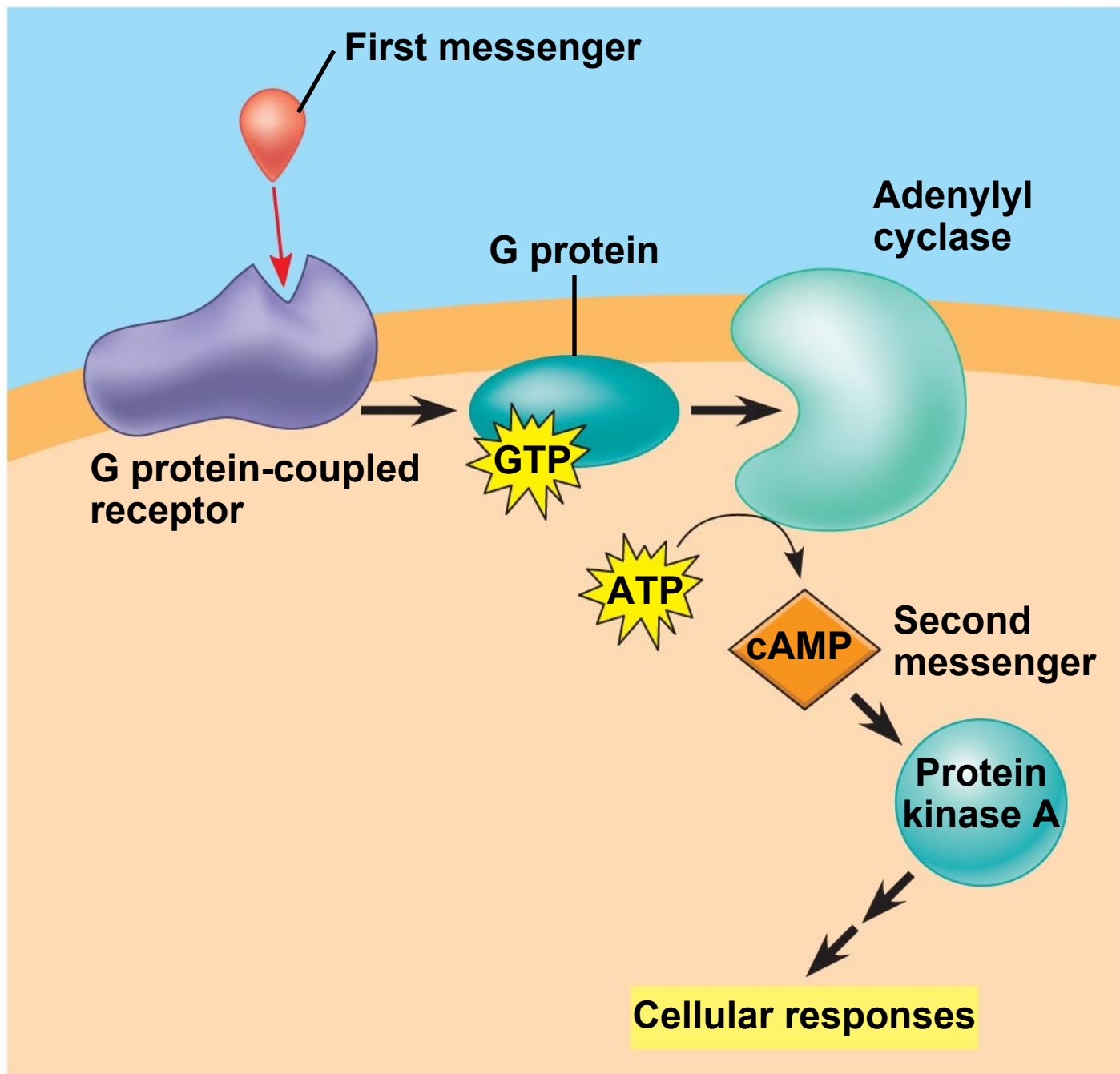
- **Cyclic AMP (cAMP)** is one of the most widely used second messengers
- **Adenylyl cyclase**, an enzyme in the plasma membrane, converts ATP to cAMP in response to an extracellular signal

Fig. 11-10



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- Many signal molecules trigger formation of cAMP
 - Other components of cAMP pathways are G proteins, G protein-coupled receptors, and protein kinases
 - cAMP usually activates protein kinase A, which phosphorylates various other proteins
 - Further regulation of cell metabolism is provided by G-protein systems that inhibit adenylyl cyclase

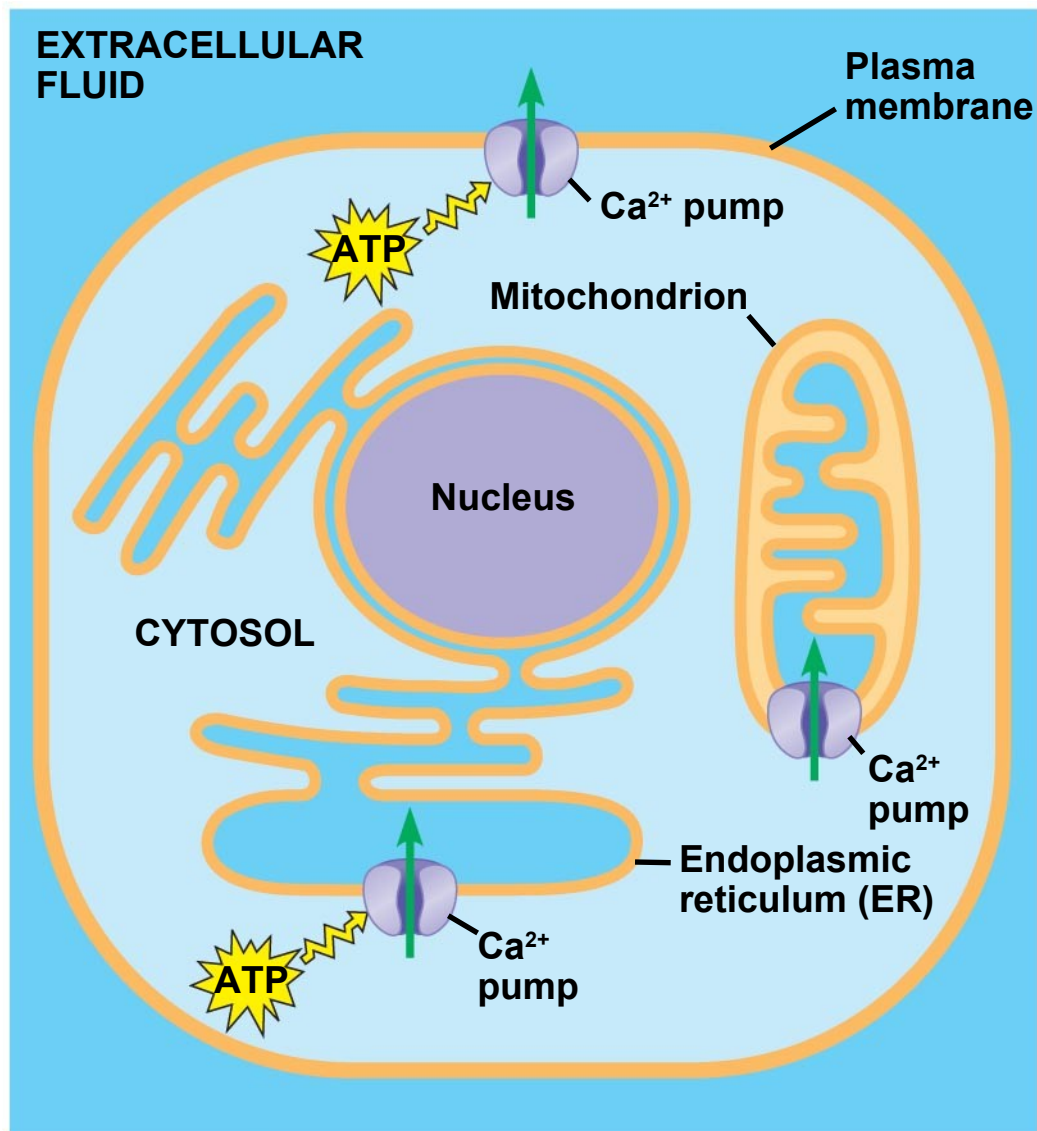
Fig. 11-11



Calcium Ions and Inositol Triphosphate (IP₃)

- Calcium ions (Ca^{2+}) act as a second messenger in many pathways
- Calcium is an important second messenger because cells can regulate its concentration

Fig. 11-12



Key

High [Ca²⁺]

Low [Ca²⁺]

-
- A signal relayed by a signal transduction pathway may trigger an increase in calcium in the cytosol
 - Pathways leading to the release of calcium involve **inositol triphosphate (IP₃)** and **diacylglycerol (DAG)** as additional second messengers

Fig. 11-13-1

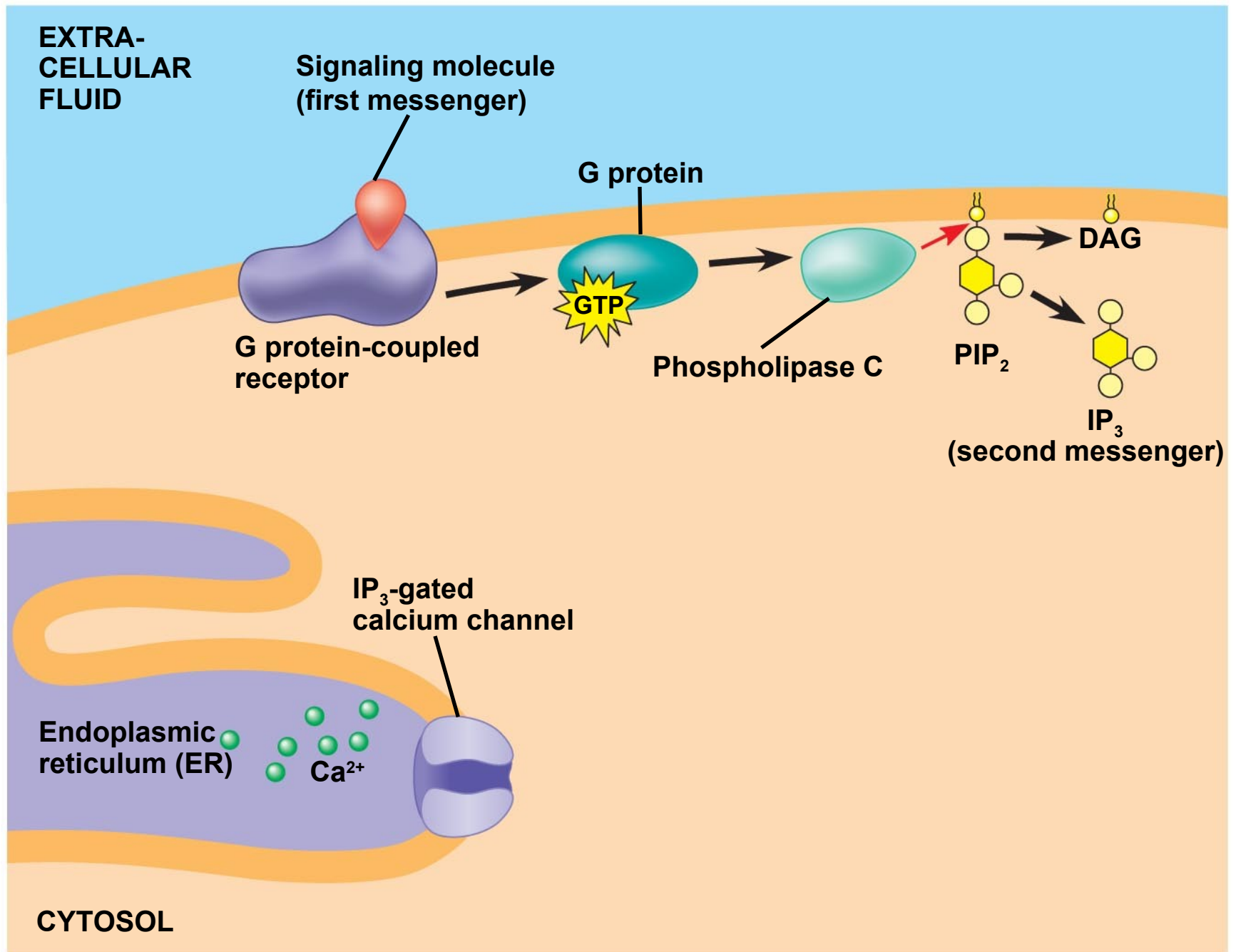


Fig. 11-13-2

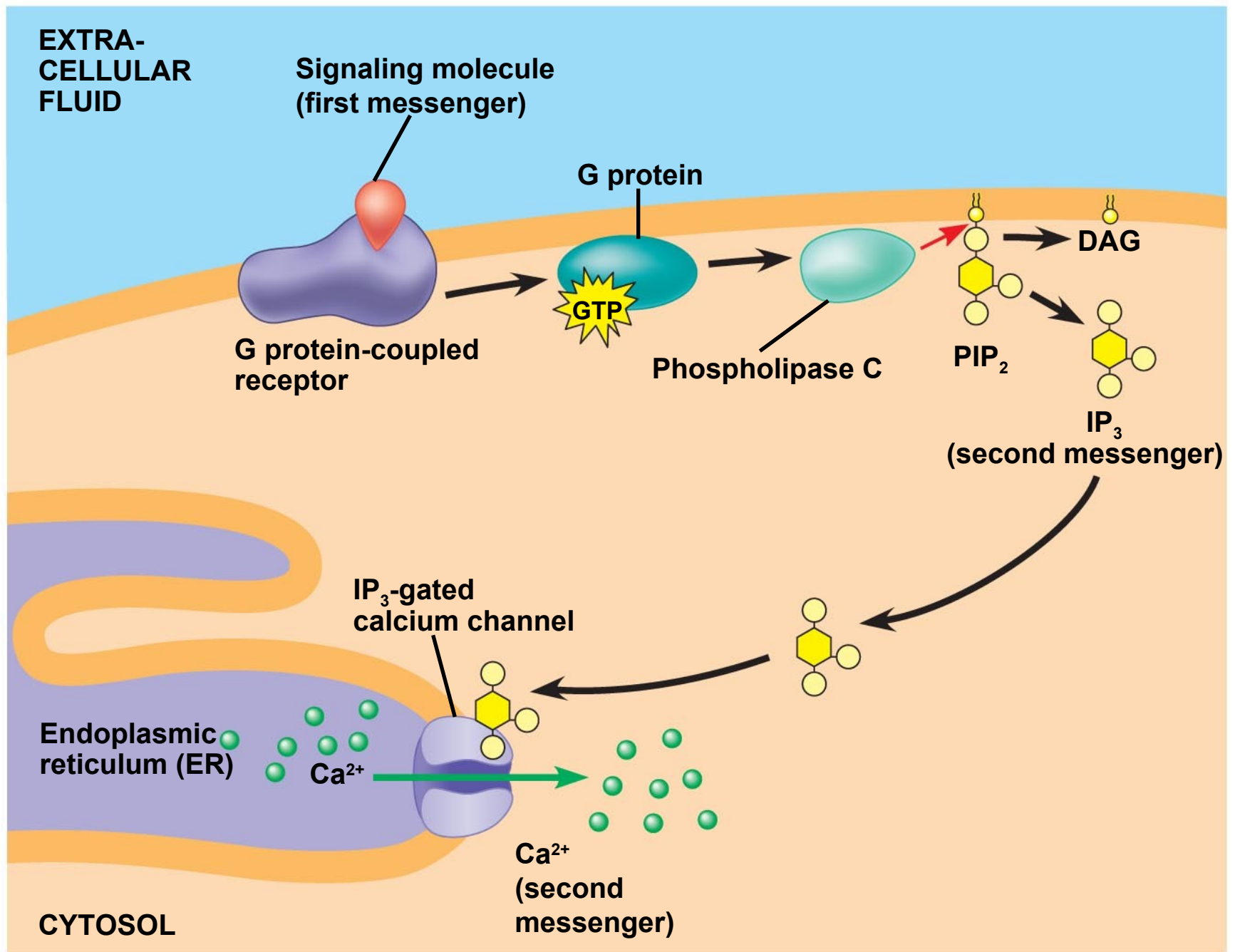
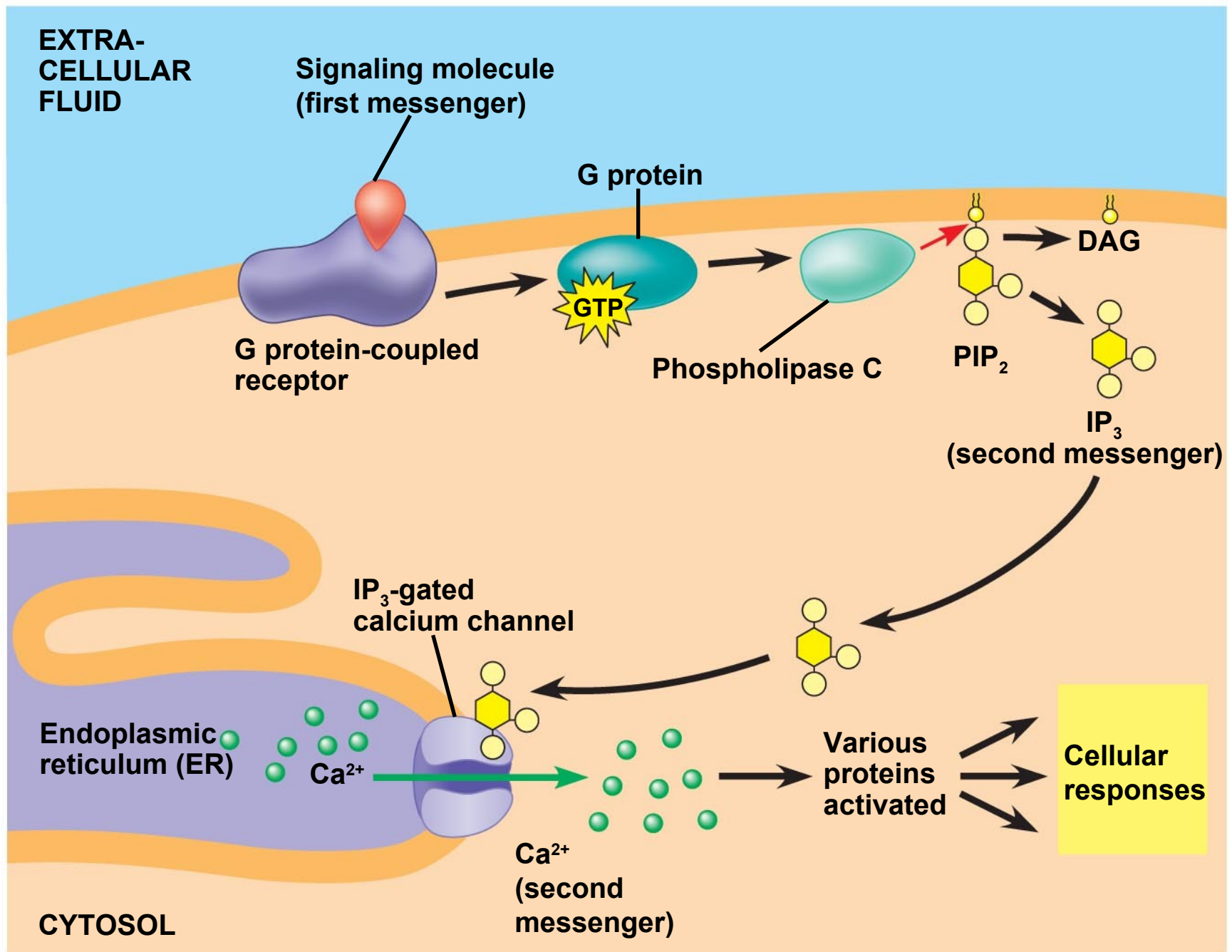


Fig. 11-13-3



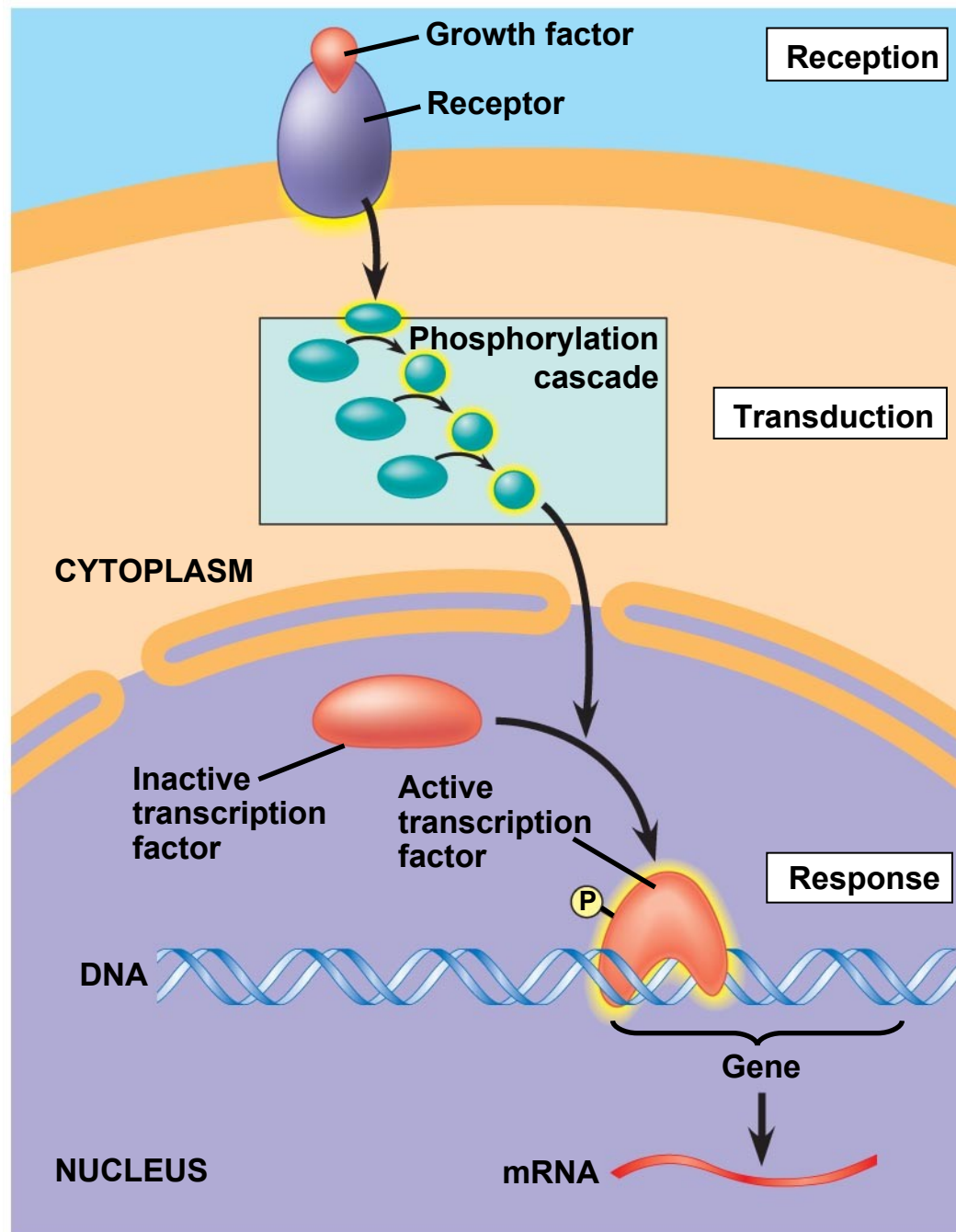
Concept 11.4: Response: Cell signaling leads to regulation of transcription or cytoplasmic activities

- The cell's response to an extracellular signal is sometimes called the “output response”

Nuclear and Cytoplasmic Responses

- Ultimately, a signal transduction pathway leads to regulation of one or more cellular activities
- The response may occur in the cytoplasm or may involve action in the nucleus
- Many signaling pathways regulate the *synthesis* of enzymes or other proteins, usually by turning genes on or off in the nucleus
- The final activated molecule may function as a transcription factor

Fig. 11-14



-
- Other pathways regulate the *activity* of enzymes

Fig. 11-15

Reception

Binding of epinephrine to G protein-coupled receptor (1 molecule)



Transduction

Inactive G protein

Active G protein (10^2 molecules)

Inactive adenylyl cyclase

Active adenylyl cyclase (10^2)

ATP

Cyclic AMP (10^4)

Inactive protein kinase A

Active protein kinase A (10^4)

Inactive phosphorylase kinase

Active phosphorylase kinase (10^5)

Inactive glycogen phosphorylase

Active glycogen phosphorylase (10^6)

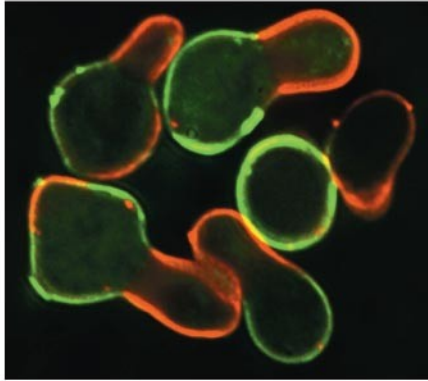
Response

Glycogen
Glucose-1-phosphate
(10^8 molecules)

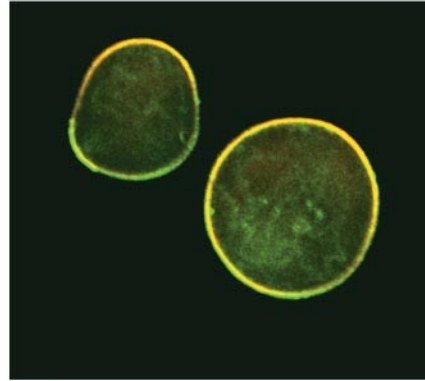
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- Signaling pathways can also affect the physical characteristics of a cell, for example, cell shape

Fig. 11-16

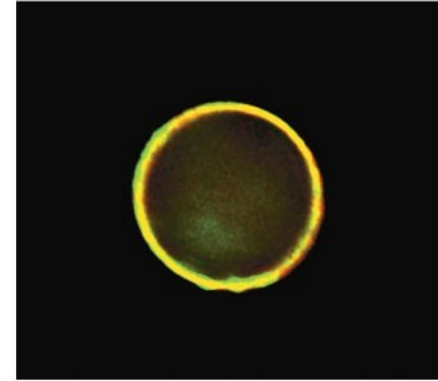
RESULTS



Wild-type (shmoos)

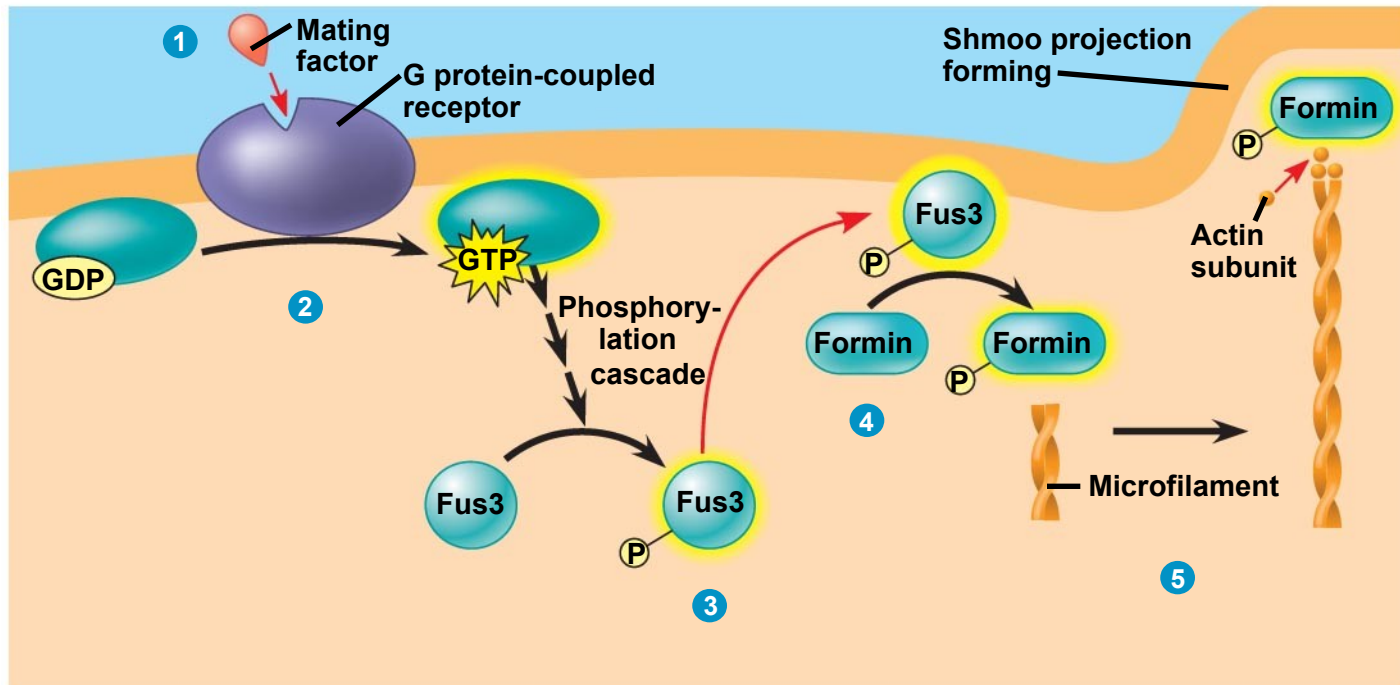


Δ Fus3



Δ formin

CONCLUSION



Fine-Tuning of the Response

- Multistep pathways have two important benefits:
 - Amplifying the signal (and thus the response)
 - Contributing to the specificity of the response

Signal Amplification

- Enzyme cascades amplify the cell's response
- At each step, the number of activated products is much greater than in the preceding step

The Specificity of Cell Signaling and Coordination of the Response

- Different kinds of cells have different collections of proteins
- These different proteins allow cells to detect and respond to different signals
- Even the same signal can have different effects in cells with different proteins and pathways
- Pathway branching and “cross-talk” further help the cell coordinate incoming signals

Fig. 11-17

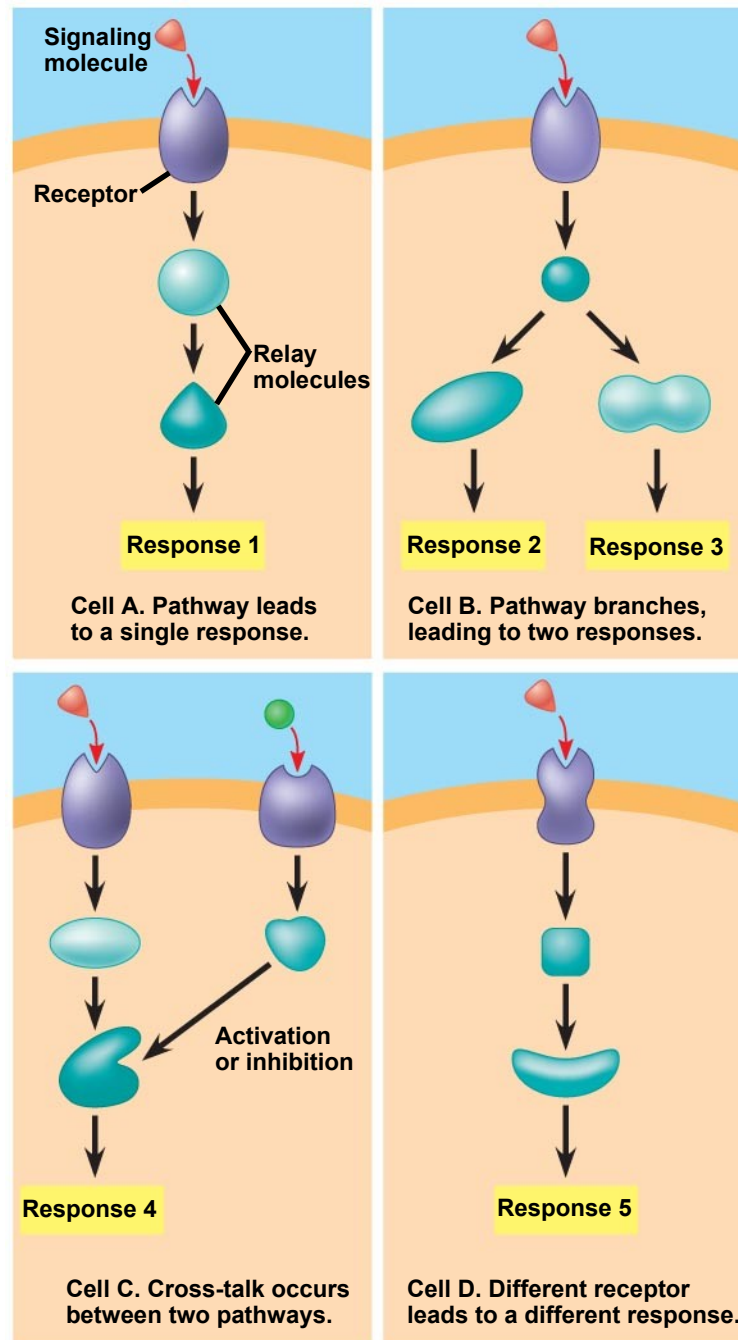


Fig. 11-17a

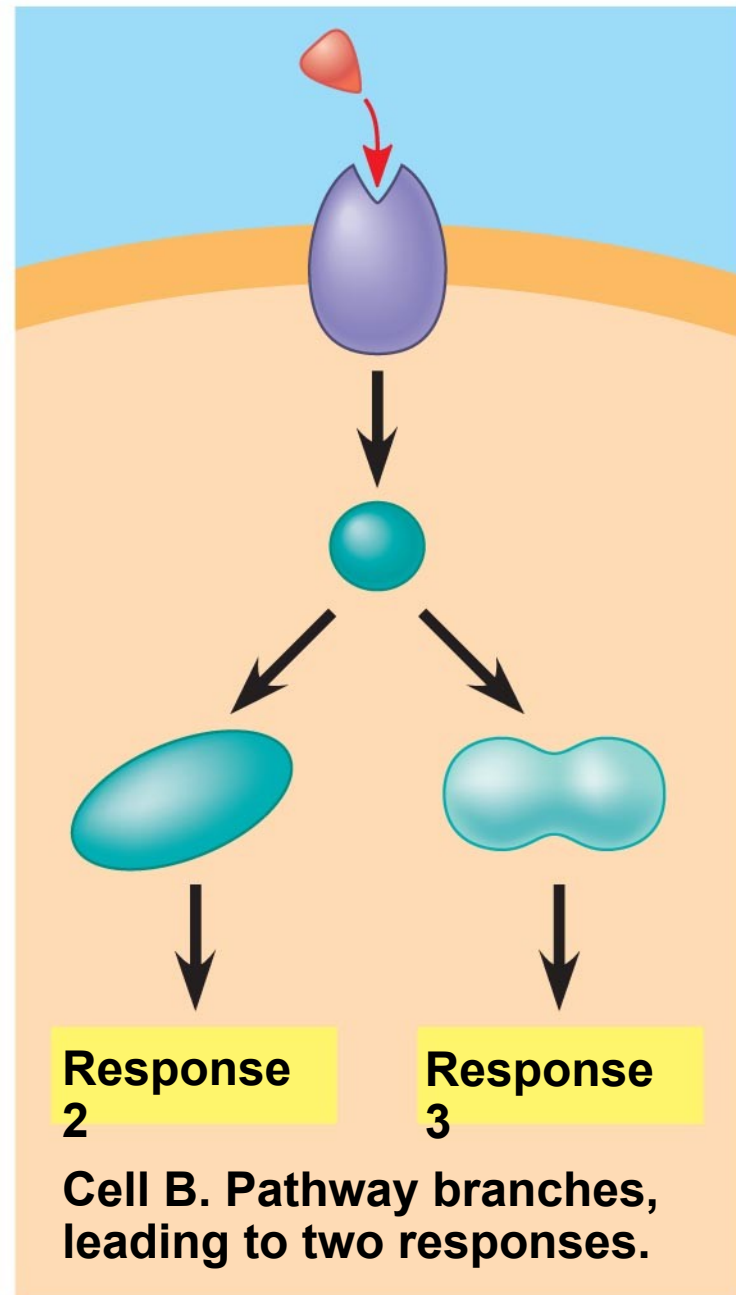
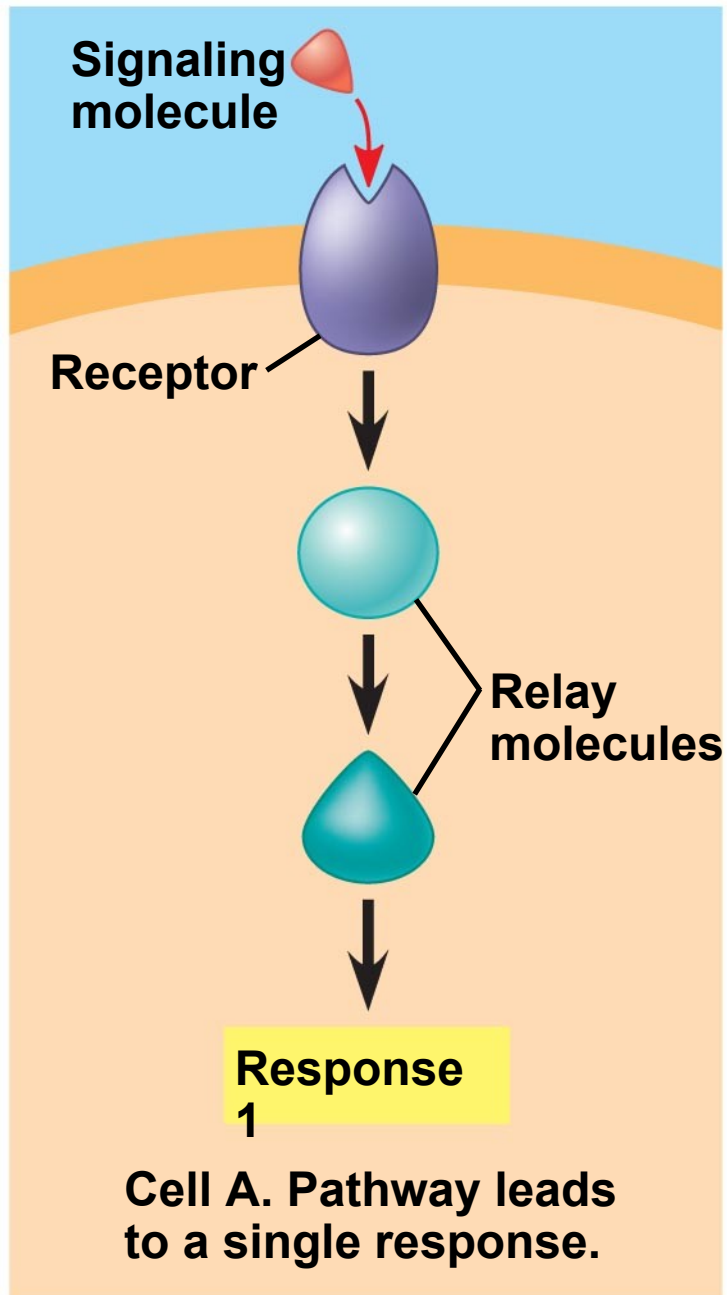
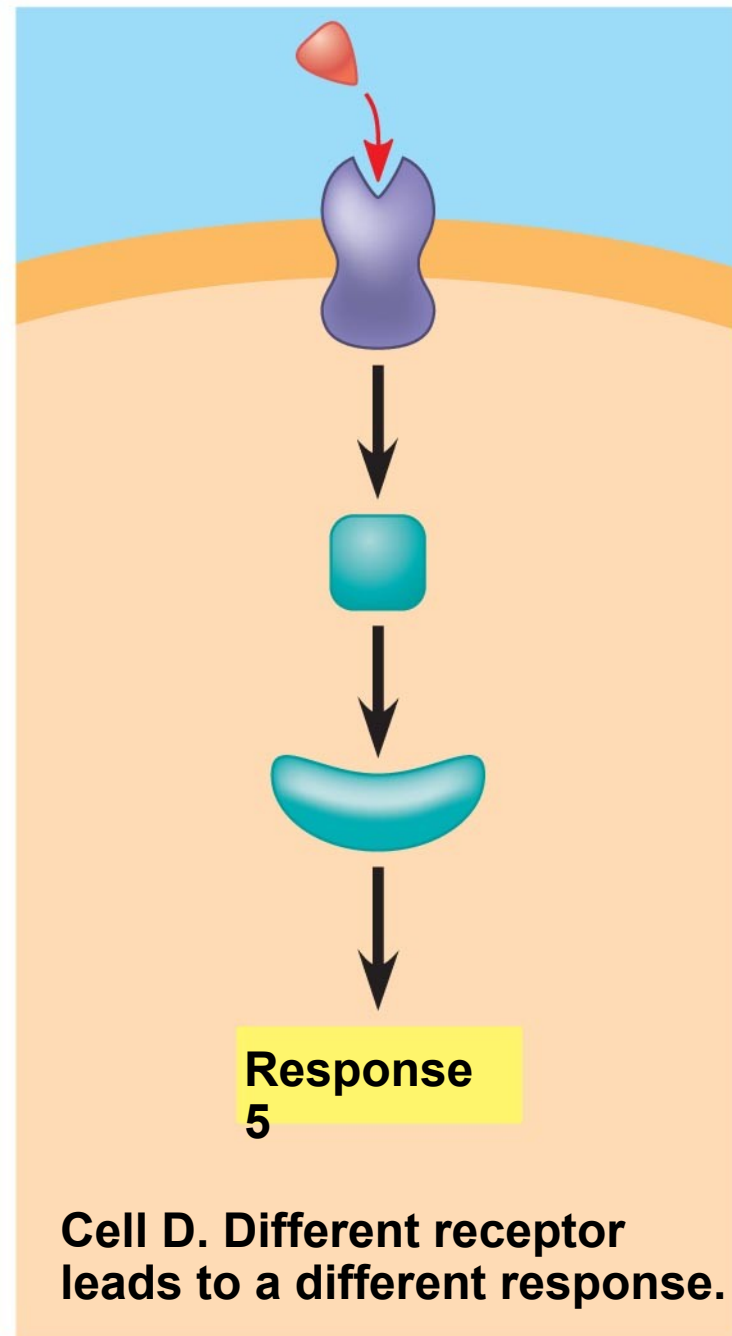
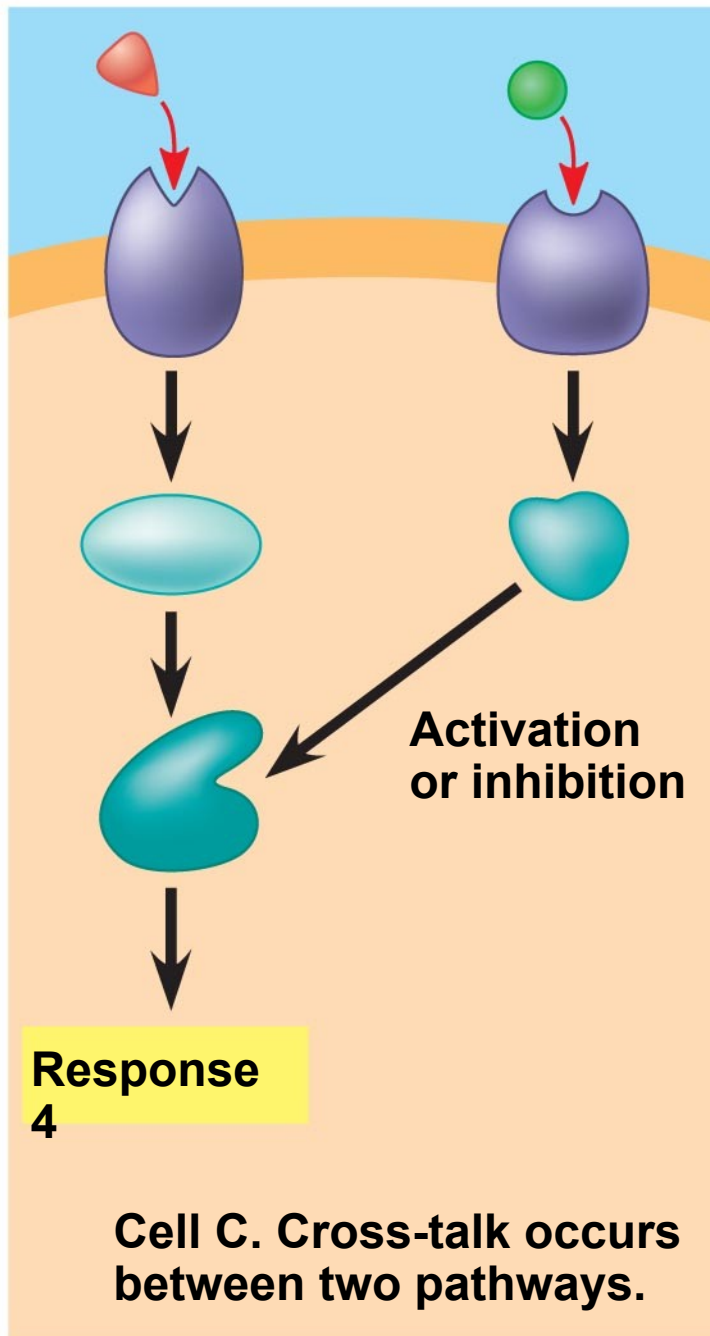


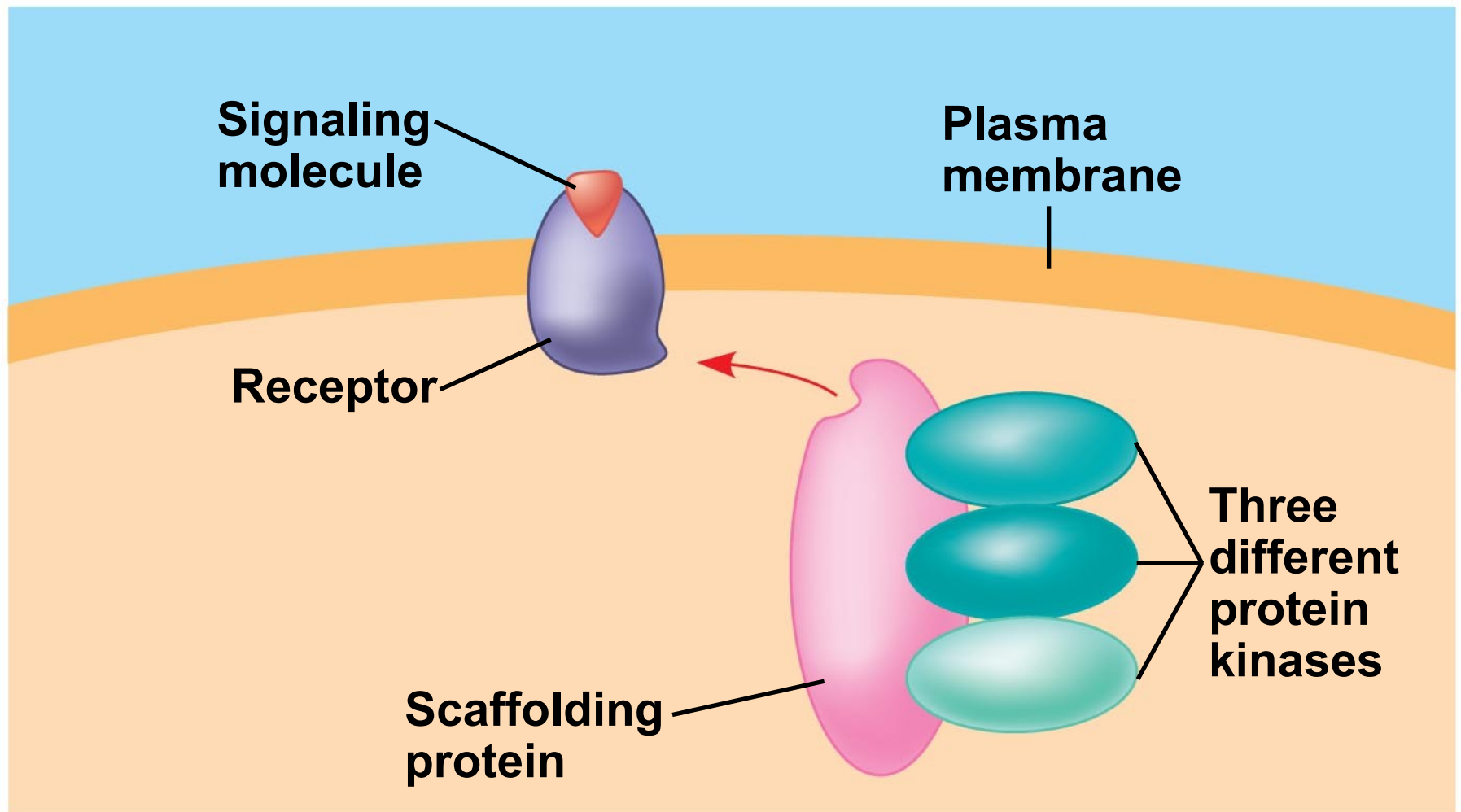
Fig. 11-17b



Signaling Efficiency: Scaffolding Proteins and Signaling Complexes

- **Scaffolding proteins** are large relay proteins to which other relay proteins are attached
- Scaffolding proteins can increase the signal transduction efficiency by grouping together different proteins involved in the same pathway

Fig. 11-18



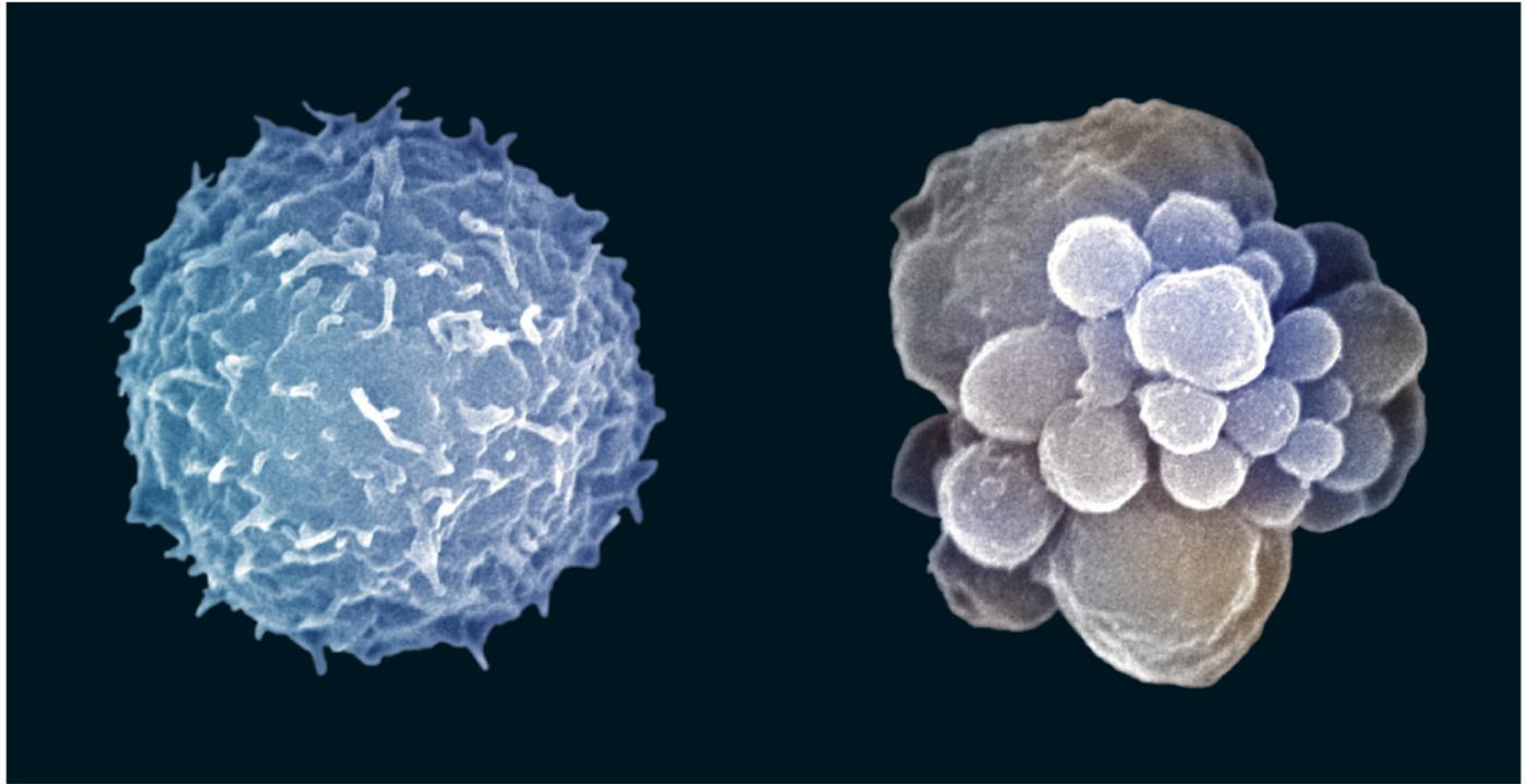
Termination of the Signal

- Inactivation mechanisms are an essential aspect of cell signaling
- When signal molecules leave the receptor, the receptor reverts to its inactive state

Concept 11.5: Apoptosis (programmed cell death) integrates multiple cell-signaling pathways

- **Apoptosis** is programmed or controlled cell suicide
- A cell is chopped and packaged into vesicles that are digested by scavenger cells
- Apoptosis prevents enzymes from leaking out of a dying cell and damaging neighboring cells

Fig. 11-19

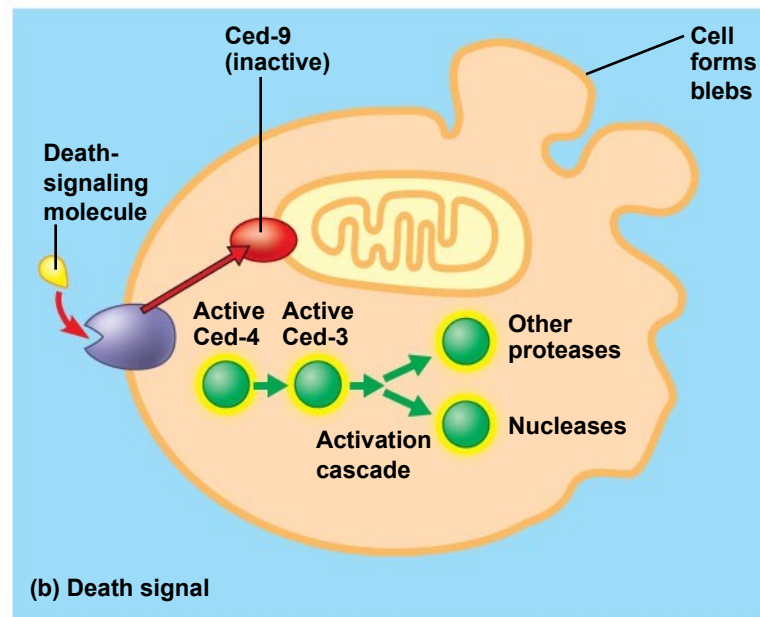
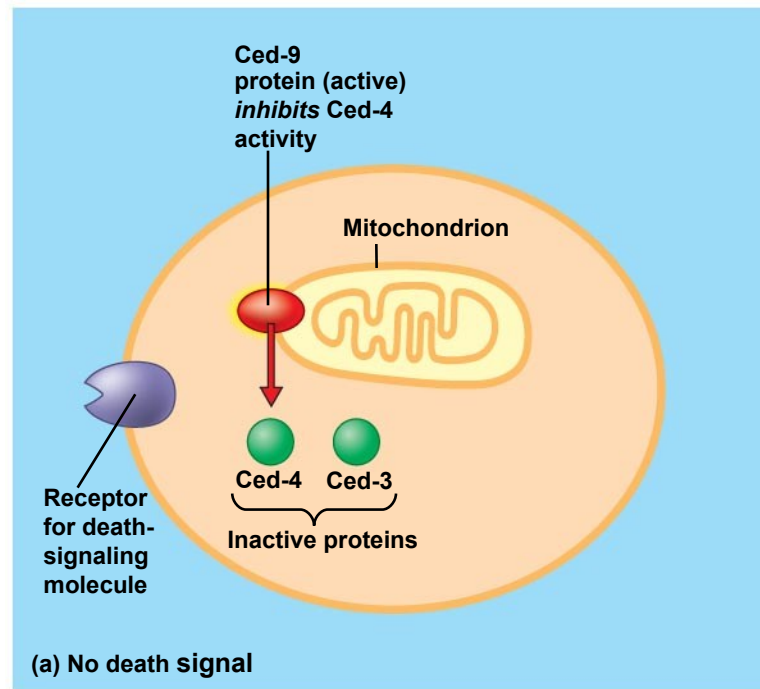


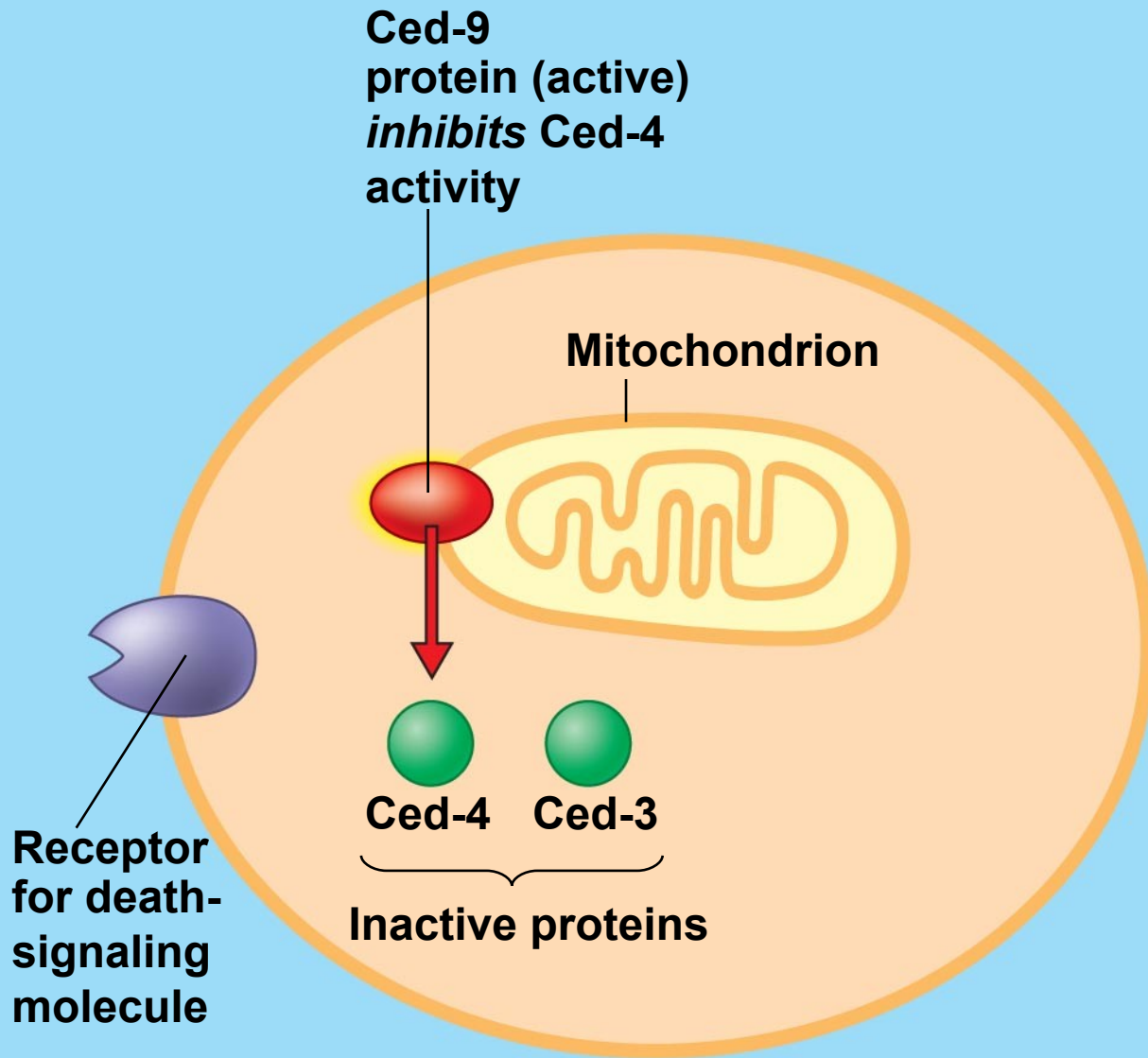
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Apoptosis in the Soil Worm *Caenorhabditis elegans*

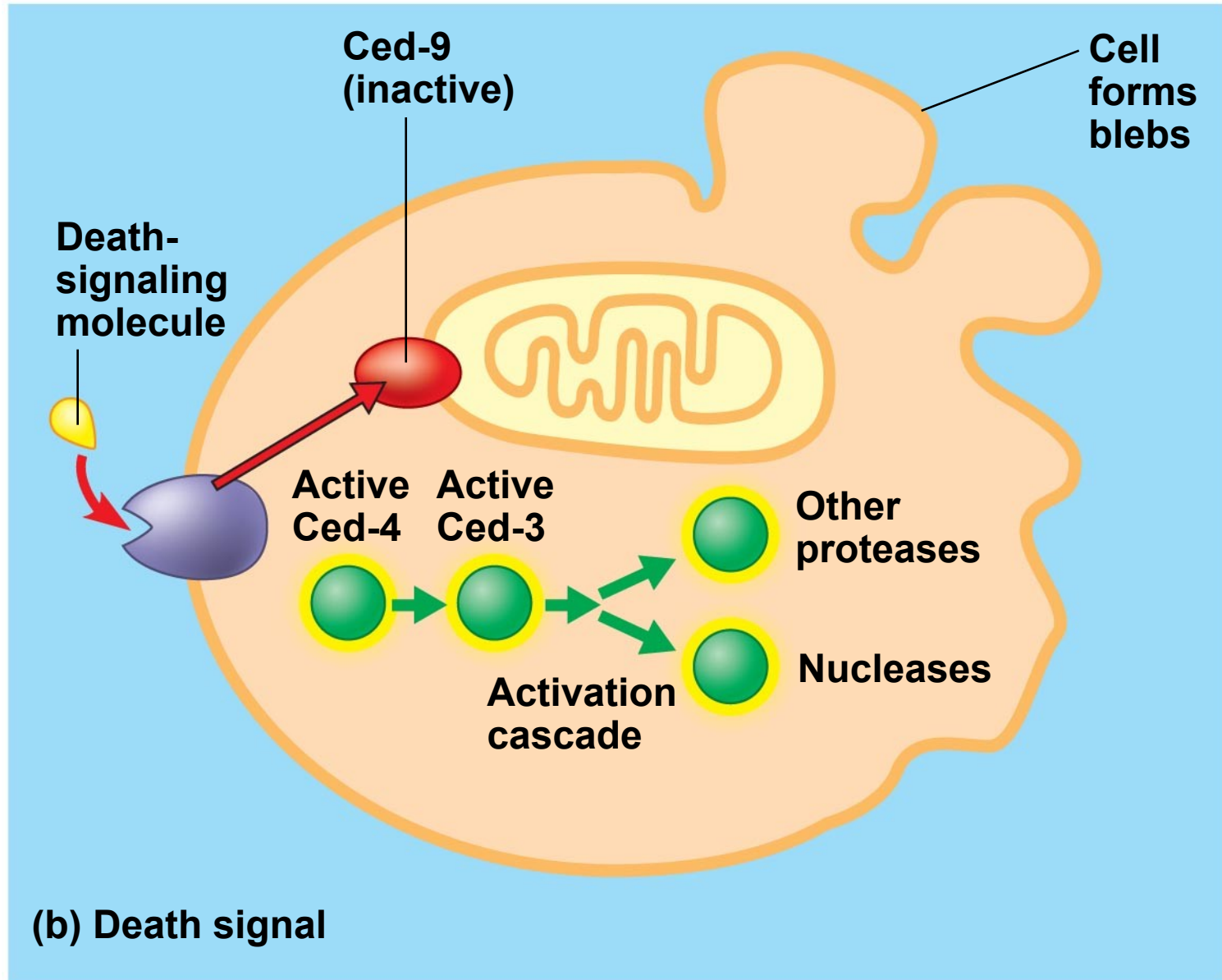
- Apoptosis is important in shaping an organism during embryonic development
- The role of apoptosis in embryonic development was first studied in *Caenorhabditis elegans*
- In *C. elegans*, apoptosis results when specific proteins that “accelerate” apoptosis override those that “put the brakes” on apoptosis

Fig. 11-20





(a) No death signal

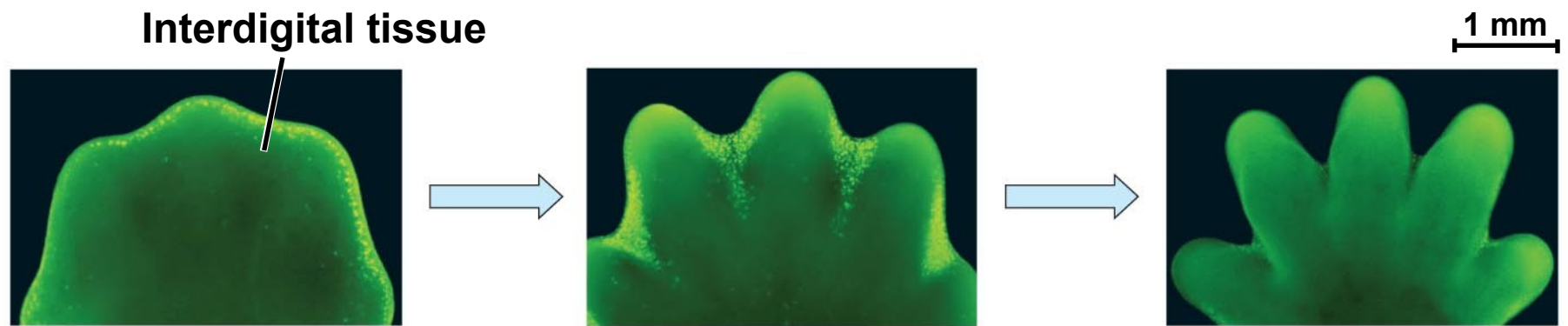


Apoptotic Pathways and the Signals That Trigger Them

- Caspases are the main proteases (enzymes that cut up proteins) that carry out apoptosis
- Apoptosis can be triggered by:
 - An extracellular death-signaling ligand
 - DNA damage in the nucleus
 - Protein misfolding in the endoplasmic reticulum

-
- Apoptosis evolved early in animal evolution and is essential for the development and maintenance of all animals
 - Apoptosis may be involved in some diseases (for example, Parkinson's and Alzheimer's); interference with apoptosis may contribute to some cancers

Fig. 11-21



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You should now be able to:

1. Describe the nature of a ligand-receptor interaction and state how such interactions initiate a signal-transduction system
2. Compare and contrast G protein-coupled receptors, tyrosine kinase receptors, and ligand-gated ion channels
3. List two advantages of a multistep pathway in the transduction stage of cell signaling
4. Explain how an original signal molecule can produce a cellular response when it may not even enter the target cell

-
5. Define the term *second messenger*; briefly describe the role of these molecules in signaling pathways
 6. Explain why different types of cells may respond differently to the same signal molecule
 7. Describe the role of apoptosis in normal development and degenerative disease in vertebrates